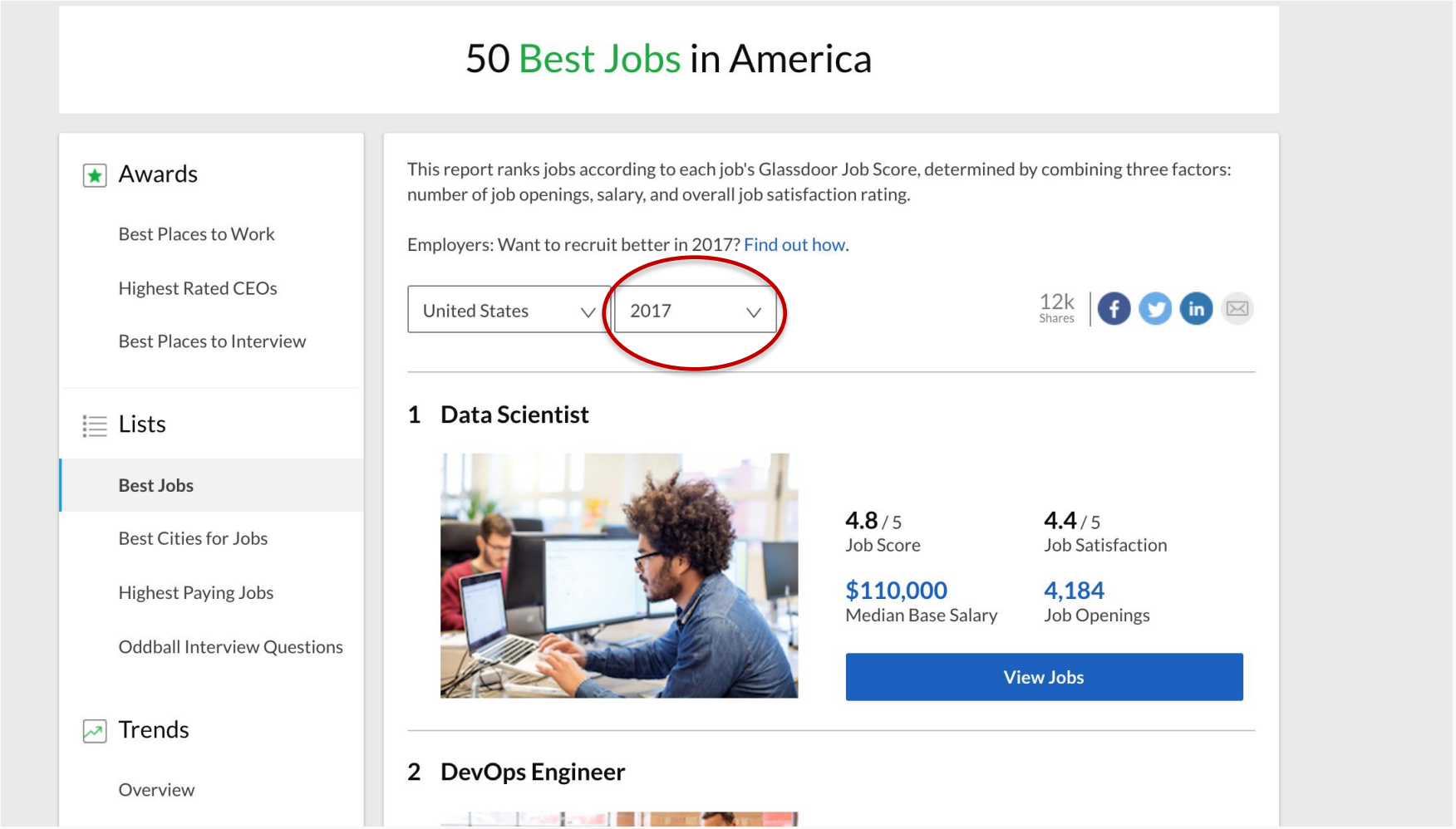
DETECTION
WWW
SOCIAL MEDIA
SERVICES
PROJECTS
BIG DATA
COMPUTING
INFORMATION
TECHNOLOGY
PROMOTION
MODELS
E-MARKETING
COMMUNICATION
MULTIMEDIA
NETWORK
CONTENT
PROCESSING
CONSUMER
ORGANIZATION
BRANDING
COMPUTER
CONSUMER DEMAND MARKETS
SOFTWARE
CODING
PREDICTIVE
PLANNING
WEB MARKETING
PROGRAM
ANALYTICS
EVENTS
PROGRAMMING
DATA MINING
DATA SCIENCE
INFORMATION
MACHINE LEARNING
VISION
ENGINEERING
RESEARCH
PROBABILITY
WEB SERVICES
BIG
DATA
SERVICE
WEB DEV
KDD
COMPUTING
MATHS PATTERN
STRATEGY
WORLDWIDE
VISUALIZATION
BIG
DATA
ENGINEERING
SOLUTIONS
PROJECTS
PLANNING
MEDIA
MOBILE
PRO
CODING
PRICING
SEGMENTATION
SOCIAL
NETWORK
STATISTICS
INFORMATION
SOCIAL NETWORKS
TARGET
DIGITAL

Lecture Outline

- Why data science?
- What is data science?
- What are Data?
- Data Explorations
- Example

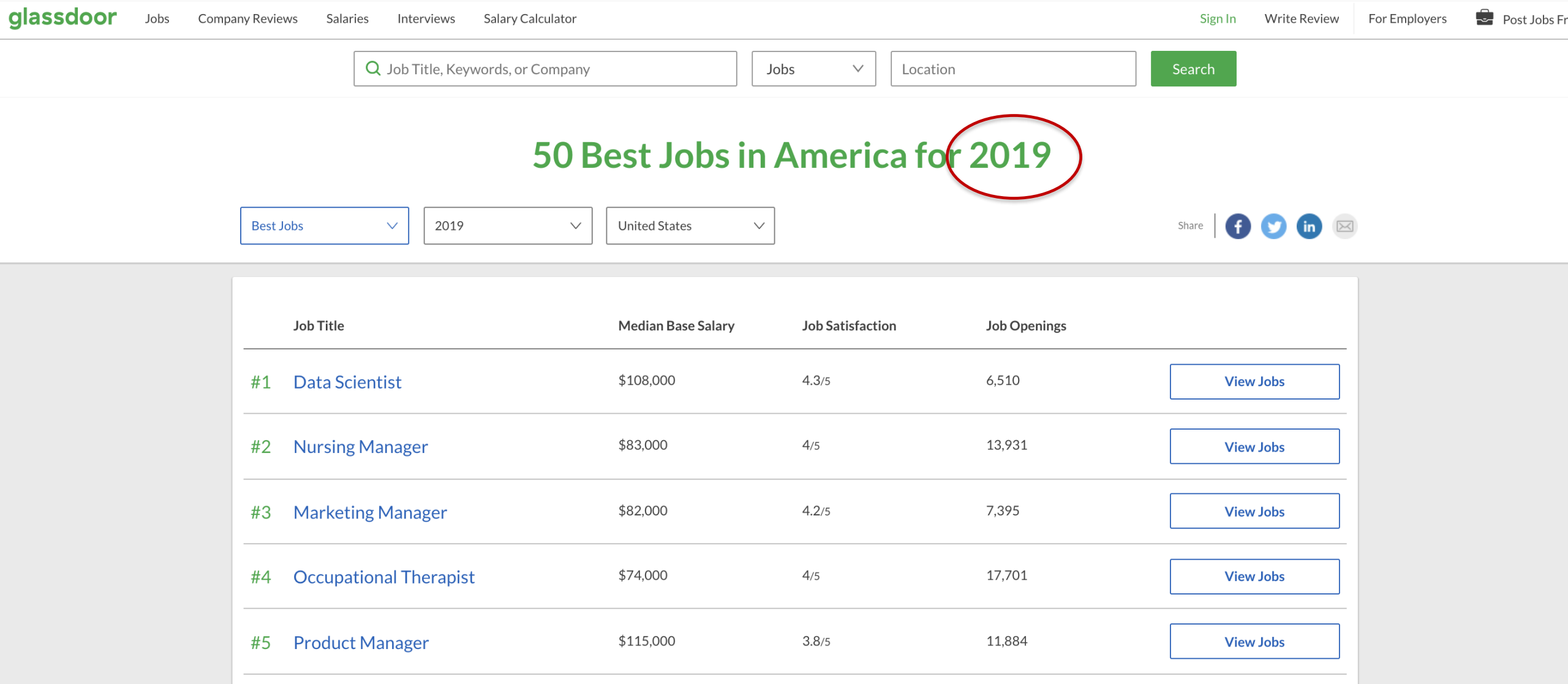
Why?

Jobs!


50 Best Jobs in America
Awards
Best Places to Work
Highest Rated CEOs
Best Places to Interview
Lists
Best Jobs
Best Cities for Jobs
Highest Paying Jobs
Oddball Interview Questions
Trends
Overview
This report ranks jobs according to each job's Glassdoor Job Score, determined by combining three factors: number of job openings, salary, and overall job satisfaction rating.
Employers: Want to recruit better in 2017? Find out how.
United States
2017
12k
Shares
1 Data Scientist
4.8 / 5
Job Score
4.4 / 5
Job Satisfaction
$110,000
Median Base Salary
4,184
Job Openings
View Jobs
2 DevOps Engineer

Why?

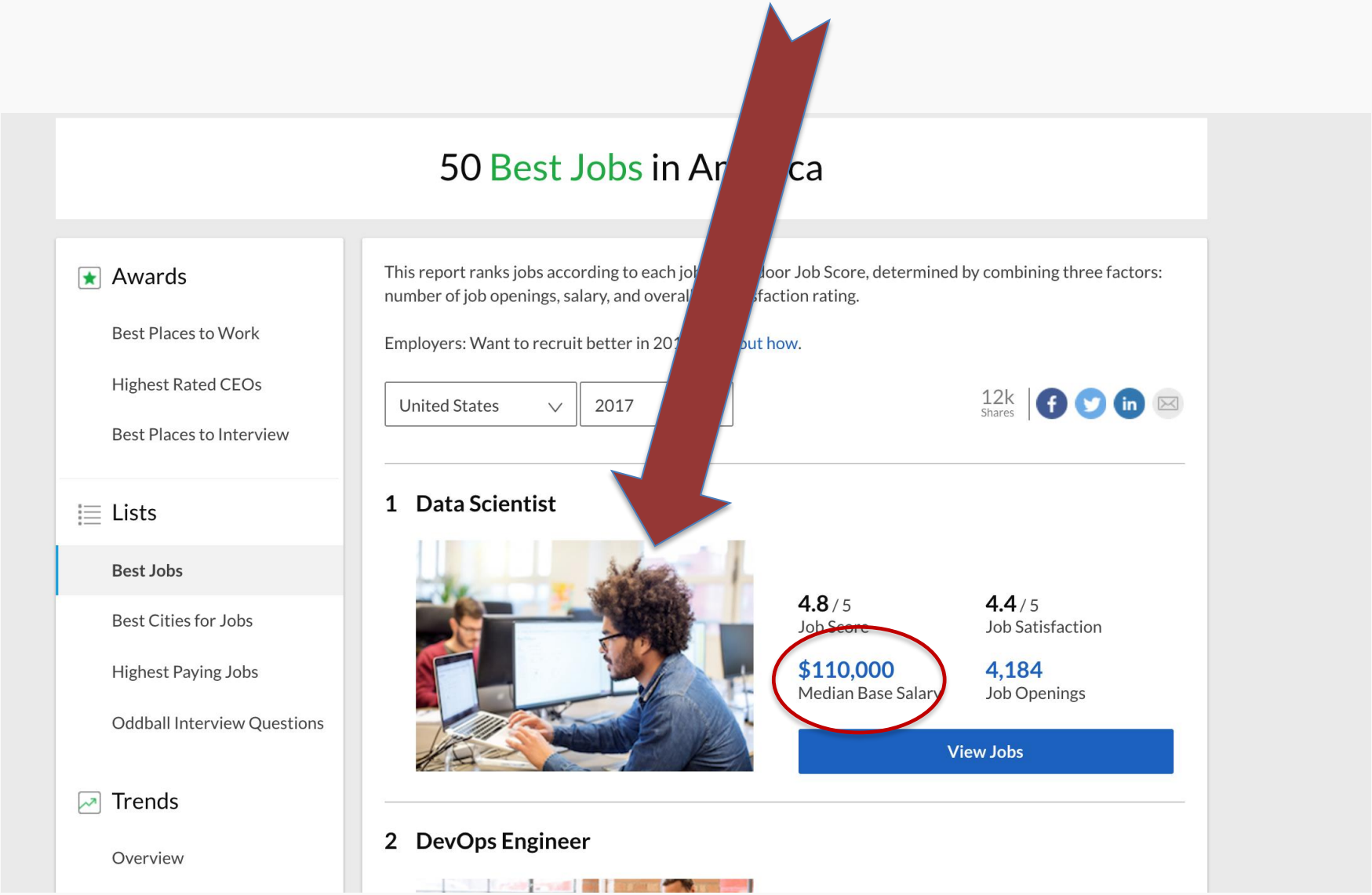
Jobs!



	Job Title	Median Base Salary	Job Satisfaction	Job Openings	
#1	Data Scientist	$108,000	4.3/5	6,510	View Jobs
#2	Nursing Manager	$83,000	4/5	13,931	View Jobs
#3	Marketing Manager	$82,000	4.2/5	7,395	View Jobs
#4	Occupational Therapist	$74,000	4/5	17,701	View Jobs
#5	Product Manager	$115,000	3.8/5	11,884	View Jobs

Why?

Jobs!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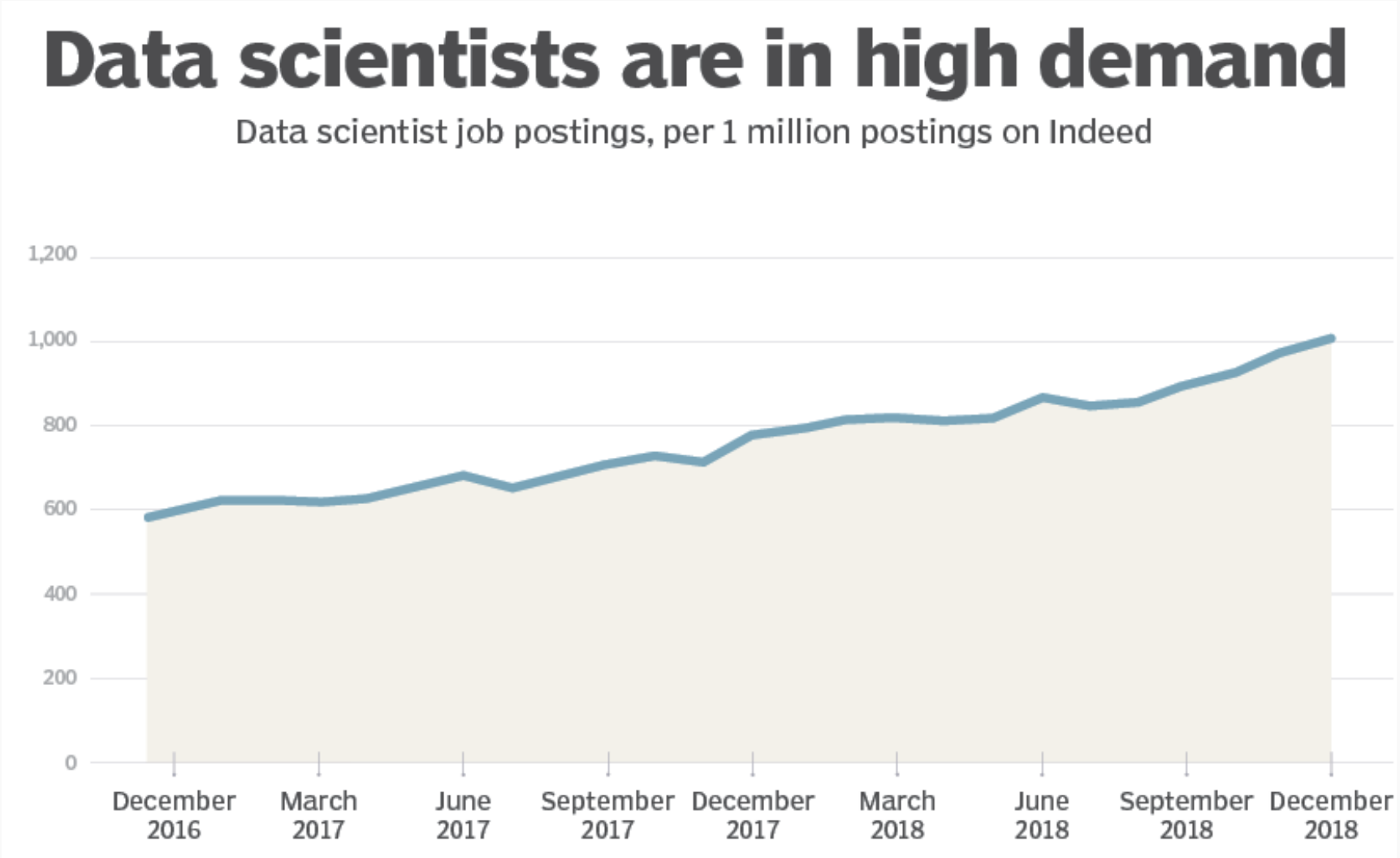
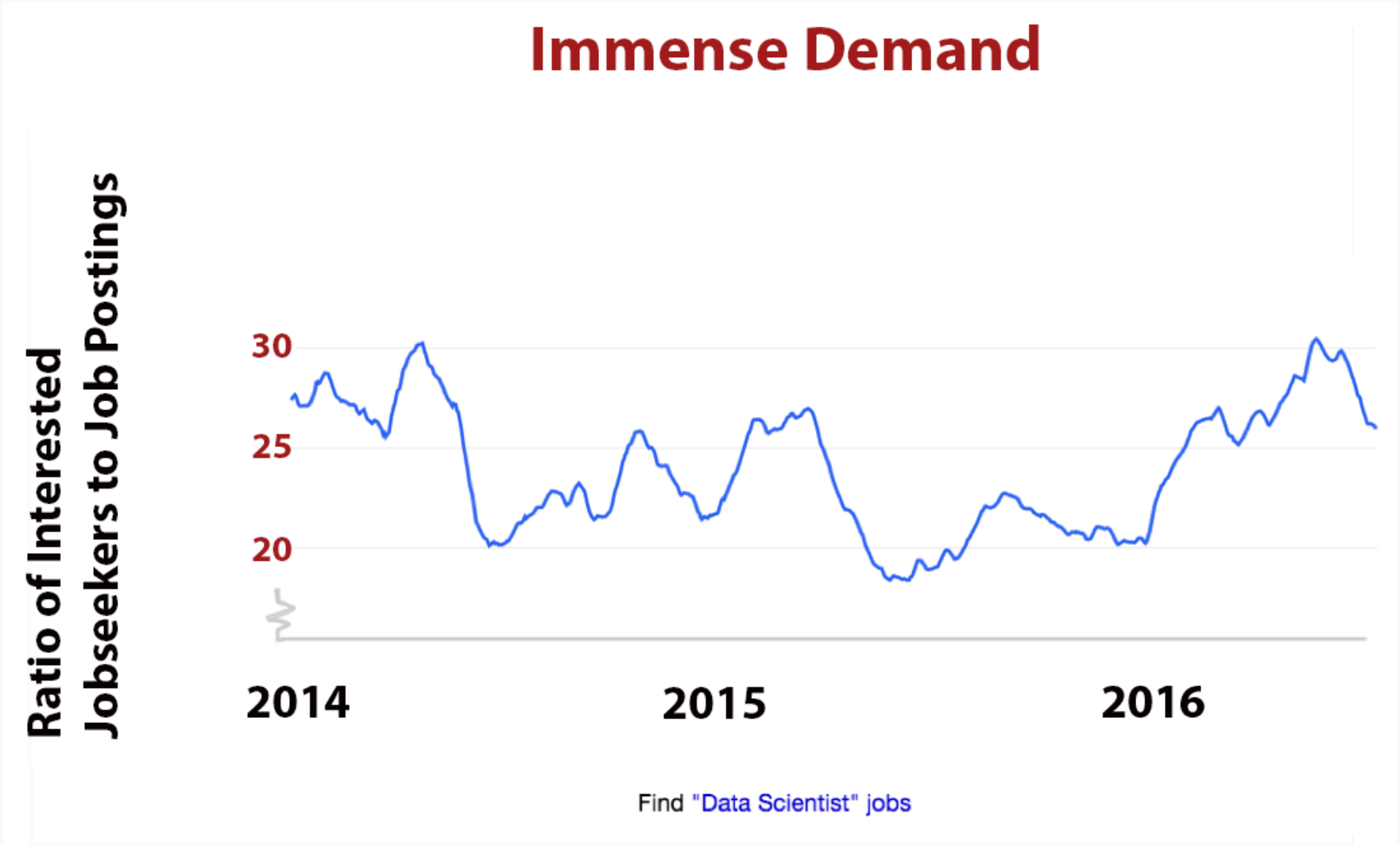
Why?


Data scientists are in high demand
Data scientist job postings, per 1 million postings on Indeed
1,200
1,000
800
600
400
200
0
December 2016
March 2017
June 2017
September 2017
December 2017
March 2018
June 2018
September 2018
December 2018

Why?


Immense Demand
Ratio of Interested Jobseekers to Job Postings
30
25
20
2014
2015
2016
Find "Data Scientist" jobs

Why?

Why do I love data science?

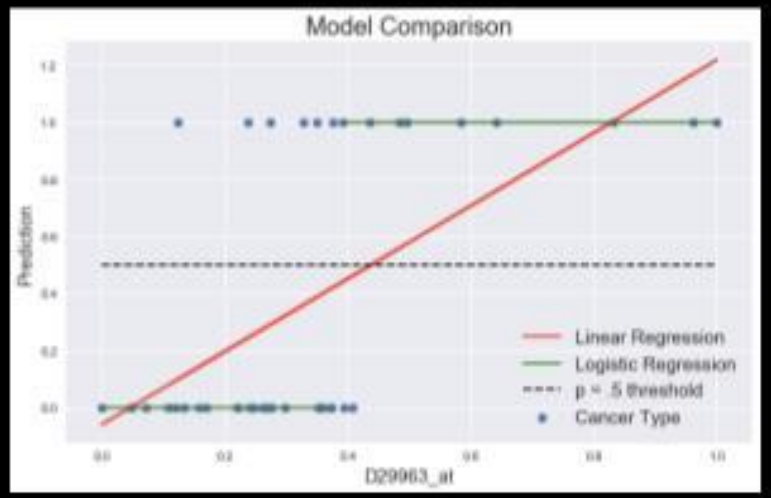
Why are you here?

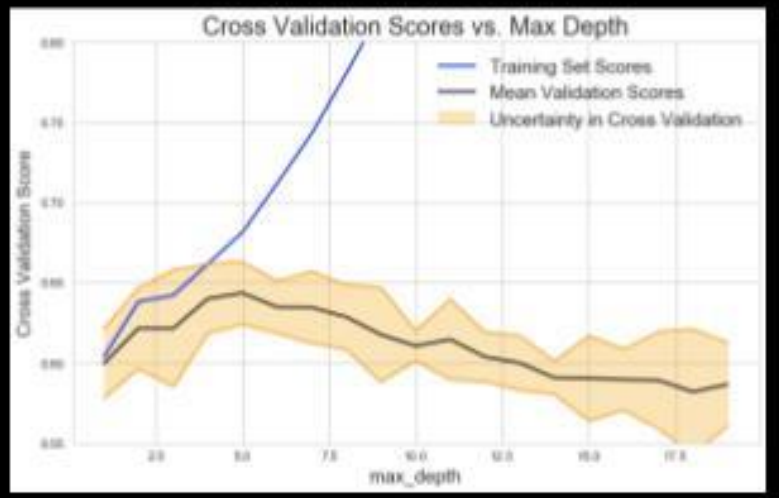
what my friends think I do

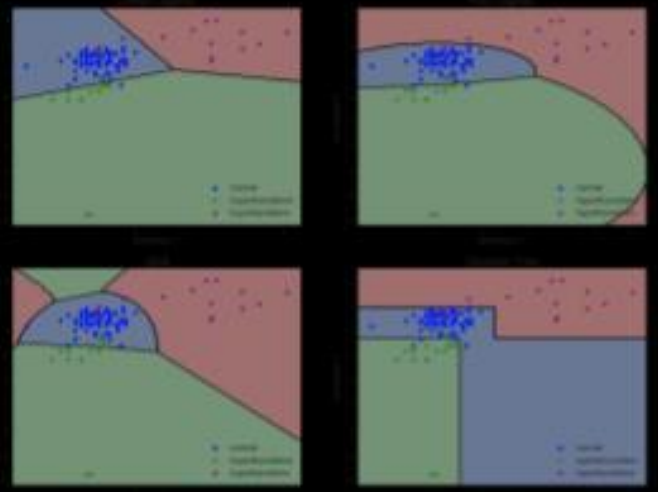
what my family thinks I do

what society thinks I do

what I actually (will) do in Data Science 1







A little bit of history

History

Long time ago (thousands of years) science was only empirical and people counted stars

History (cont)

Long time ago (thousands of years) science was only empirical and people counted stars or crops

History (cont)

Long time ago (thousands of years) science was only empirical and people counted stars or crops and used the data to create machines to describe the phenomena

History (cont)

Few hundred years: theoretical approaches, try to derive equations to describe general phenomena.

1. $\nabla \cdot \mathbf{D} = \rho_V$
2. $\nabla \cdot \mathbf{B} = 0$
3. $\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$
4. $\nabla \times \mathbf{H} = \frac{\partial \mathbf{D}}{\partial t} + \mathbf{J}$

$$T^2 = \frac{4\pi^2}{GM} a^3$$

can be expressed as simply

$$T^2 = a^3$$

If expressed in the following units:

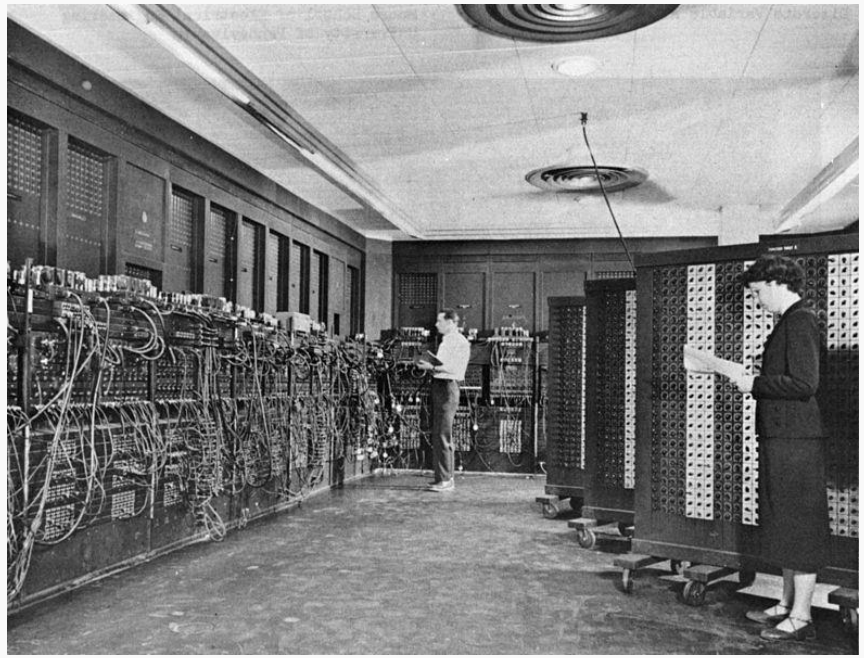
- T Earth years
- a Astronomical units AU (a = 1 AU for Earth)
- M Solar masses $M_\odot$

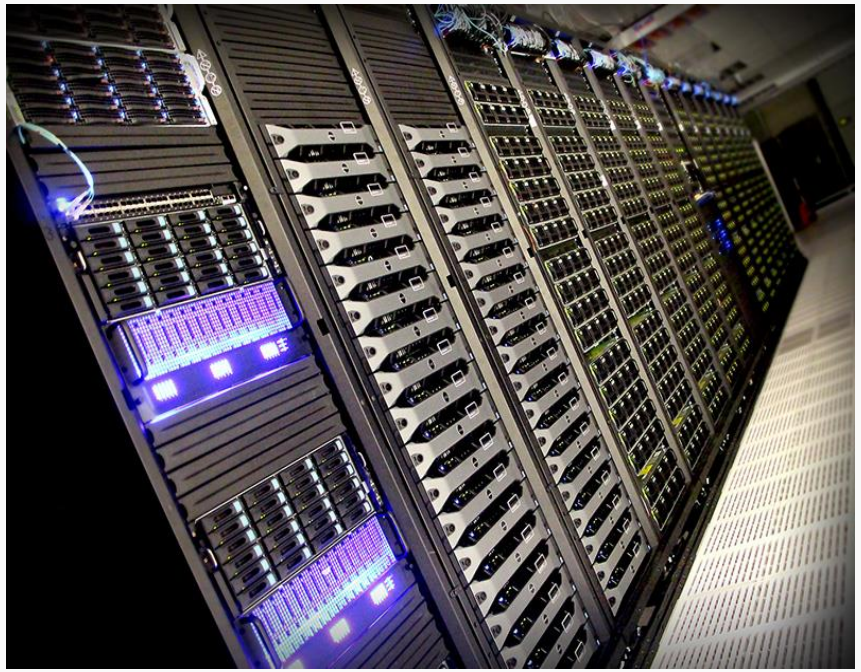
Then $\frac{4\pi^2}{G} = 1$

$$H(t)|\psi(t)\rangle = i\hbar \frac{\partial}{\partial t}|\psi(t)\rangle$$

History (cont)

About a hundred years ago: computational approaches





History (cont)

And then …. data science

- Data science is the study of data **to extract meaningful insights** for business.

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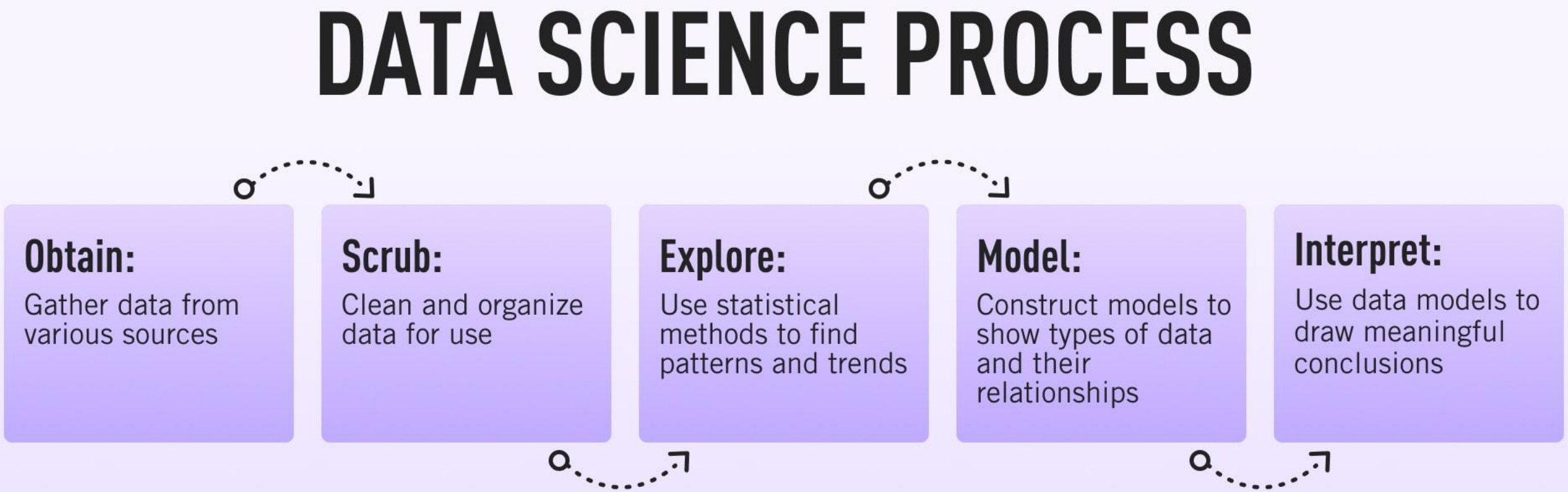
- It is a multidisciplinary approach that combines principles and practices from the fields of **mathematics**, **statistics**, **artificial intelligence**, and **computer engineering** to analyse large amounts of data.

- Data science is the study of data **to extract meaningful insights** for business.
- It is a multidisciplinary approach that combines principles and practices from the fields of **mathematics**, **statistics**, **artificial intelligence**, and **computer engineering** to analyse large amounts of data.
- This analysis helps data scientists to ask and answer questions like **what happened**, **why it happened**, what will happen, and **what can be done** with the results.

- “**Data science**, also known as **data-driven science**, is an interdisciplinary field of **scientific methods, processes, algorithms and systems to extract knowledge** or insights from data in various forms, either structured or unstructured, similar to **data mining**.”

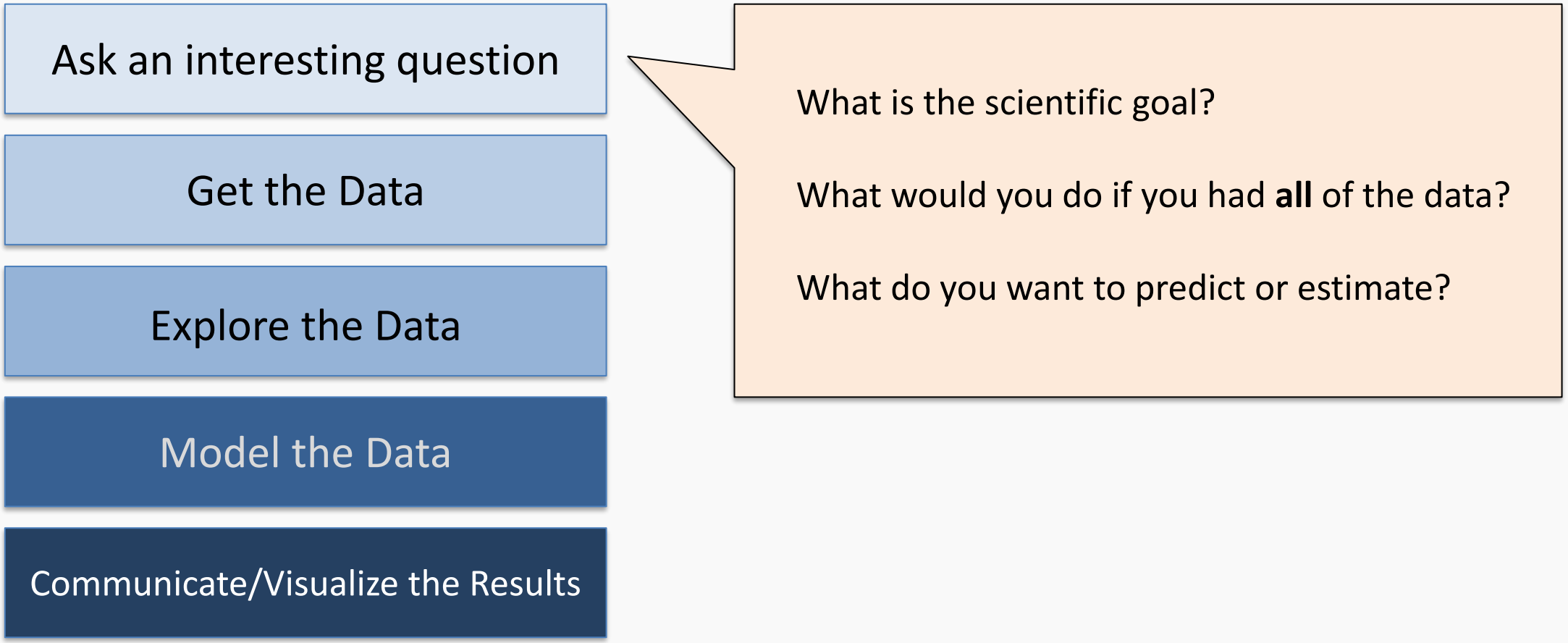
Data Science

What? The Data Science Process



What?

The Data Science Process



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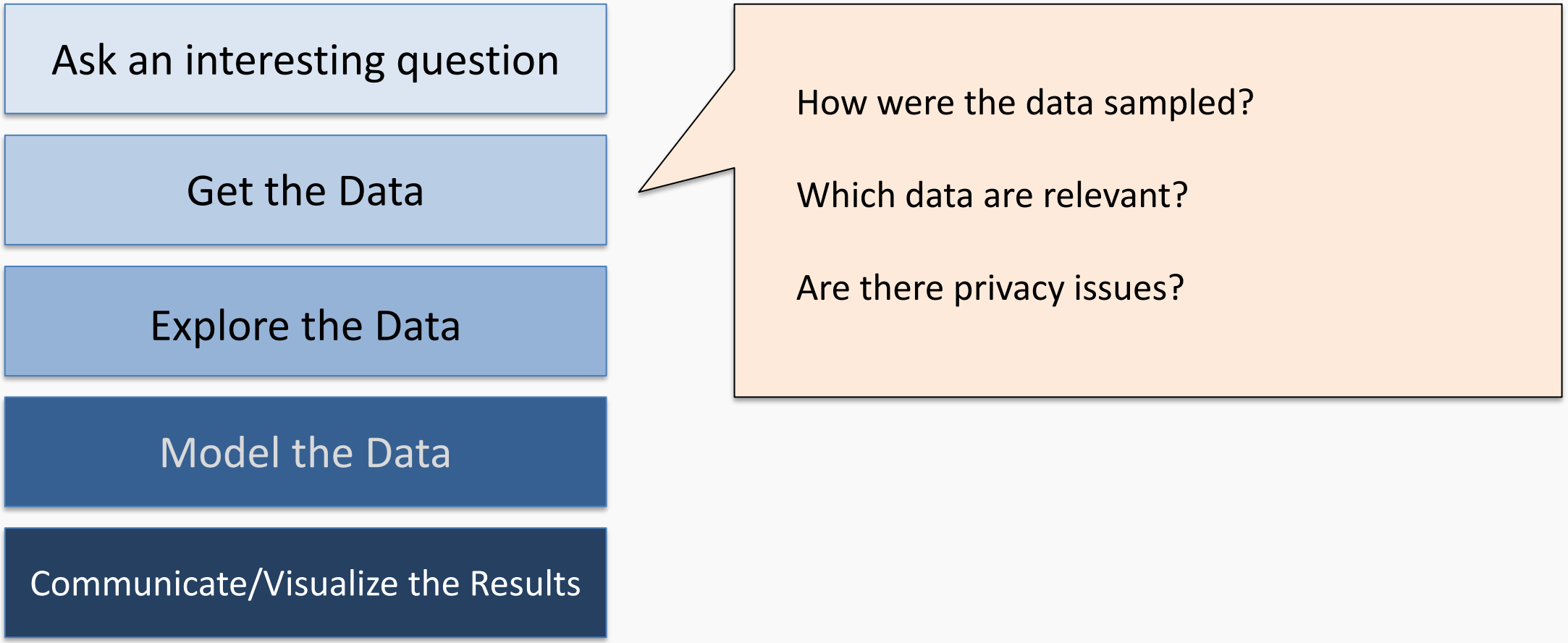

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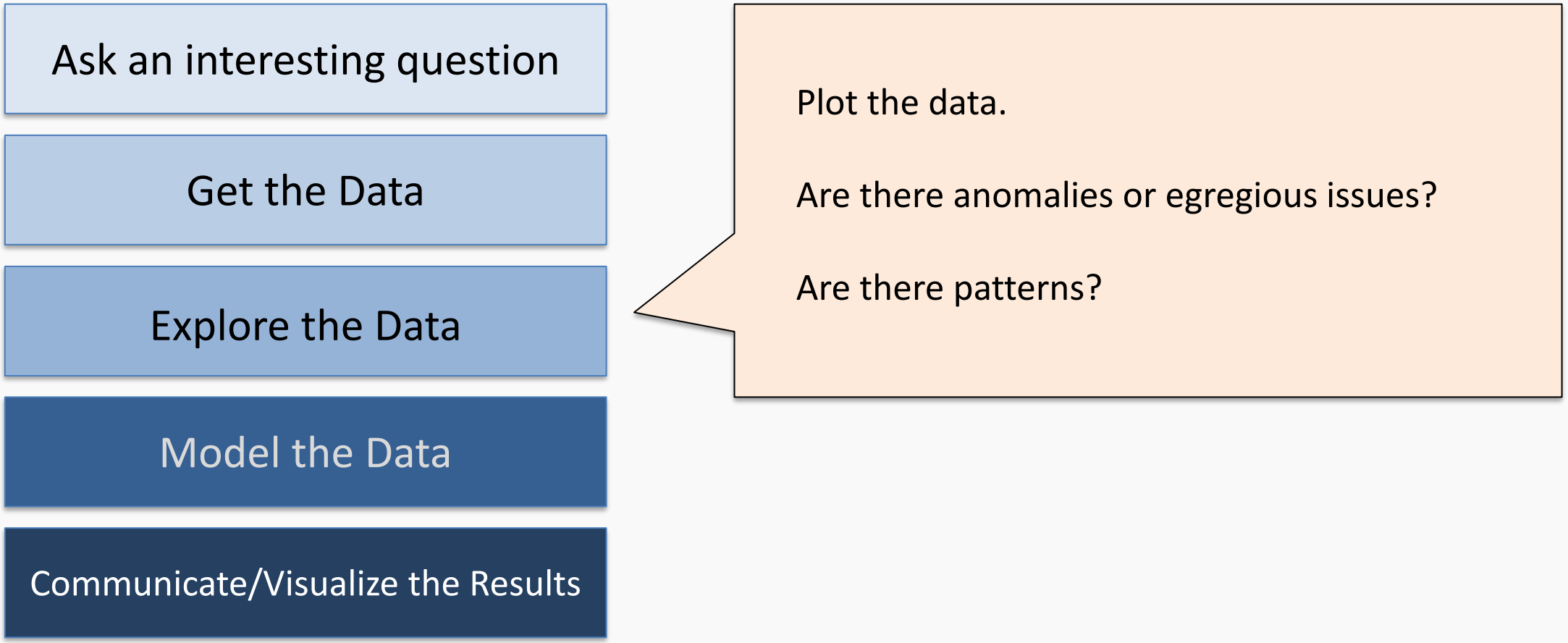

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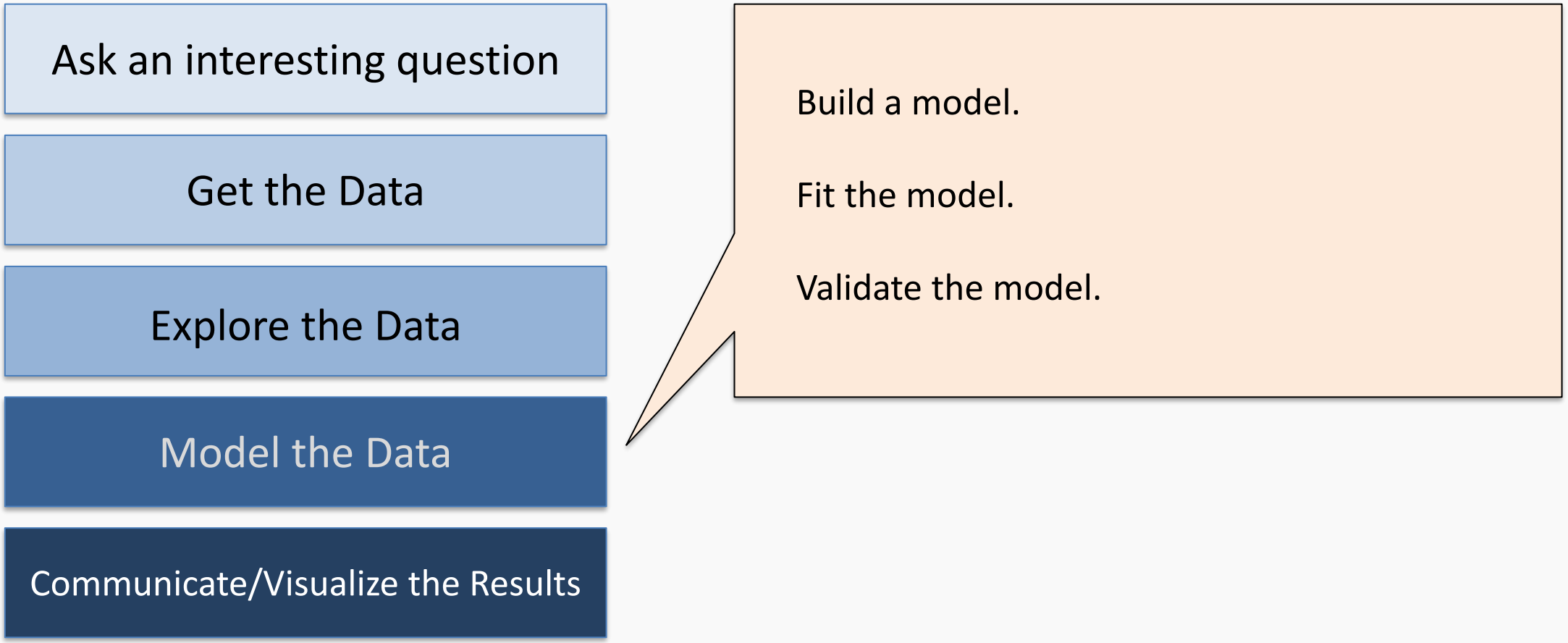
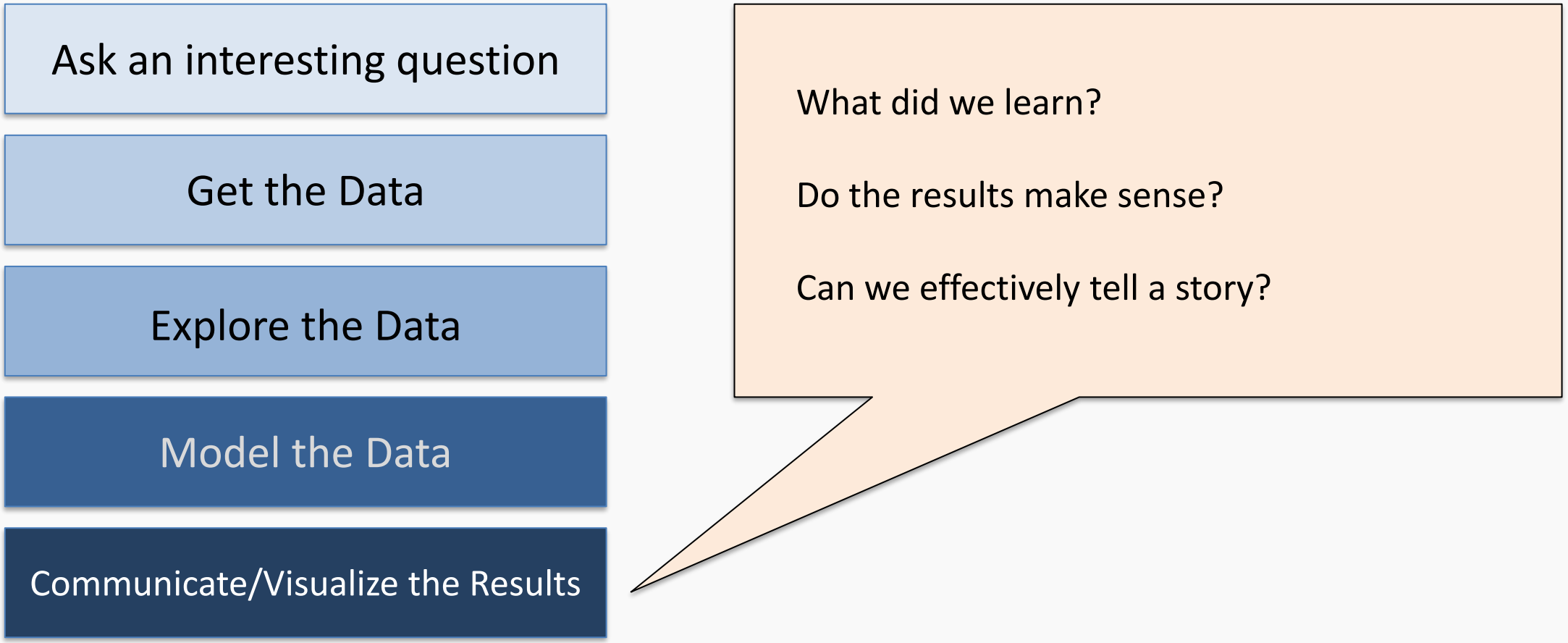


What?

The Data Science Process



What are data?

"A **datum** is a **single measurement of something** on a scale that is understandable to both the **recorder** and the **reader**. Data are multiple such measurements."

Claim: everything is (can be) data!

Where do data come from?

- **Internal sources:** already collected by or is part of the overall data collection of you organization.
  For example: business-centric data that is available in the organization data base to record day to day operations; scientific or experimental data.
- **Existing External Sources:** available in ready to read format from an outside source for free or for a fee.
  For example: public government databases, stock market data, Yelp reviews, [your favorite sport]-reference.
- **External Sources Requiring Collection Efforts:** available from external source but acquisition requires special processing.
  For example: data appearing only in print form, or data on websites.

Ways to gather online data

How to get data generated, published or hosted online:

- **API (Application Programming Interface):** using a prebuilt set of functions developed by a company to access their services. Often pay to use. For example: Google Map API, Facebook API, Twitter API
- **RSS (Rich Site Summary):** summarizes frequently updated online content in standard format. Free to read if the site has one. For example: news-related sites, blogs
- **Web scraping:** using software, scripts or by-hand extracting data from what is displayed on a page or what is contained in the HTML file (often in tables).

Web scraping

- **Why do it?**
  - Older government or smaller news sites might not have APIs for accessing data, or publish RSS feeds or have databases for download. Or, you don't want to pay to use the API or the database.
- **How do you do it?** See HW1 (beautifulsoup)
- **Should you do it?**
  - You just want to explore: Are you violating their terms of service? Privacy concerns for website and their clients?
  - You want to publish your analysis or product: Do they have an API or fee that you are bypassing? Are they willing to share this data? Are you violating their terms of service? Are there privacy concerns?

Types of data

What kind of values are in your data (data types)?

Simple or atomic:

- **Numeric:** integers, floats
- **Boolean:** binary or true false values
- **Strings:** sequence of symbols

Data types

What kind of values are in your data (data types)? Compound, composed of a bunch of atomic types:

- **Date and time:** compound value with a specific structure
- **Lists:** a list is a sequence of values
- **Dictionaries:** A dictionary is a collection of key-value pairs, a pair of values *x* : *y* where *x* is usually a string called the key representing the “name” of the entry, and *y* is a value of any type.

Example: Student record: what are *x* and *y*?

Data storage

How is your data represented and stored (data format)?

- **Tabular Data:** a dataset that is a two-dimensional table, where each row typically represents a single data record, and each column represents one type of measurement (csv, dat, xlsx, etc.).
- **Structured Data:** each data record is presented in a form of a [possibly complex and multi-tiered] dictionary (json, xml, etc.)
- **Semi structured Data:** not all records are represented by the same set of keys or some data records are not represented using the key-value pair structure.

Tabular Data

In tabular data, we expect each record or observation to represent a set of measurements of a single object or event.

First Look At The Data

In [27]:
```
hubway_data = pd.read_csv('hubway_trips.csv', low_memory=False)
hubway_data.head()
```

Out[27]:

	seq_id	hubway_id	status	duration	start_date	strt_statn	end_date	end_statn	bike_nr	subsc_type	zip_code	birth_da
0	1	8	Closed	9	7/28/2011 10:12:00	23.0	7/28/2011 10:12:00	23.0	B00468	Registered	'97217	1976.0
1	2	9	Closed	220	7/28/2011 10:21:00	23.0	7/28/2011 10:25:00	23.0	B00554	Registered	'02215	1966.0
2	3	10	Closed	56	7/28/2011 10:33:00	23.0	7/28/2011 10:34:00	23.0	B00456	Registered	'02108	1943.0
3	4	11	Closed	64	7/28/2011 10:35:00	23.0	7/28/2011 10:36:00	23.0	B00554	Registered	'02116	1981.0
4	5	12	Closed	12	7/28/2011 10:37:00	23.0	7/28/2011 10:37:00	23.0	B00554	Registered	'97214	1983.0

Tabular Data

First Look At The Data

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2	3	10	Closed	56	7/28/2011 10:33:00	23.0	7/28/2011 10:34:00	23.0	B00456	Registered	'02108	1943.0
3	4	11	Closed	64	7/28/2011 10:35:00	23.0	7/28/2011 10:36:00	23.0	B00554	Registered	'02116	1981.0
4	5	12	Closed	12	7/28/2011 10:37:00	23.0	7/28/2011 10:37:00	23.0	B00554	Registered	'97214	1983.0

- Each type of measurement is called a **variable** or an **attribute** of the data (e.g. `seq_id`, `status` and `duration` are variables or attributes). The number of attributes is called the **dimension**. These are often called **features**.
- We expect each table to contain a set of **records** or **observations** of the same kind of object or event (e.g. our table above contains observations of rides/checkouts).

Types of Data

We'll see later that it's important to distinguish between classes of variables or attributes based on the type of values they can take on.

- **Quantitative variable:** is numerical and can be either:
  - **discrete -** a finite number of values are possible in any bounded interval. For example: "Number of siblings" is a discrete variable
  - **continuous -** an infinite number of values are possible in any bounded interval. For example: "Height" is a continuous variable
- **Categorical variable:** no inherent order among the values For example: "What kind of pet you have" is a categorical variable

Common Issues

Common issues with data:

- **Missing values:** how do we fill in?
- **Wrong values:** how can we detect and correct?
- **Messy format**
- **Not usable:** the data cannot answer the question posed

Messy Data

The following is a table accounting for the number of produce deliveries over a weekend.

What are the variables in this dataset? What object or event are we measuring?

	Friday	Saturday	Sunday
Morning	15	158	10
Afternoon	2	90	20
Evening	55	12	45

What's the issue? How do we fix it?

Messy Data

We’re measuring individual deliveries; the variables are Time, Day, Number of Produce.

	Friday	Saturday	Sunday
Morning	15	158	10
Afternoon	2	90	20
Evening	55	12	45

Problem: each column header represents a single value rather than a variable. Row headers are “hiding” the Day variable. The values of the variable, “Number of Produce”, is not recorded in a single column.

Fixing Messy Data

We need to reorganize the information to make explicit the event we're observing and the variables associated to this event.

ID	Time	Day	Number
1	Morning	Friday	15
2	Morning	Saturday	158
3	Morning	Sunday	10
4	Afternoon	Friday	2
5	Afternoon	Saturday	9
6	Afternoon	Sunday	20
7	Evening	Friday	55
8	Evening	Saturday	12
9	Evening	Sunday	45

Tabular = Happy Pavlos ☺

Common causes of messiness are:

- Column headers are values, not variable names
- Variables are stored in both rows and columns
- Multiple variables are stored in one column/entry
- Multiple types of experimental units stored in same table

In general, we want each file to correspond to a dataset, each column to represent a single variable and each row to represent a single observation.

We want to **tabularize** the data. This makes Python happy.

Lecture Outline

- Why data science?
- What is data science?
- What are Data?
- **Data Explorations**
- Example

Data Exploration: Descriptive Statistics

Basics of Sampling

Population versus sample:

- A **population** is the entire set of objects or events under study. Population can be hypothetical “all students” or all students in this class.
- A **sample** is a “representative” subset of the objects or events under study. Needed because it’s impossible or intractable to obtain or compute with population data.

Biases in samples:

- **Selection bias**: some subjects or records are more likely to be selected
- Volunteer/**nonresponse bias**: subjects or records who are not easily available are not represented

Examples?

Sample mean

- The mean is the most common **measure of the location** of a set of points.
- However, the mean is very **sensitive to outliers**.

$$\text{mean}(x) = \overline{x} = \frac{1}{m}\sum_{i=1}^{m} x_i$$

Sample median

The **median** of a set of n number of observations in a sample, ordered by value, of a variable is is defined by

$$\text{Median} = \begin{cases} x_{(n+1)/2} & \text{if } n \text{ is odd} \\ \frac{x_{n/2} + x_{(n+1)/2}}{2} & \text{if } n \text{ is even} \end{cases}$$

Example (already in order):

Ages: 17, 19, 21, 22, 23, 23, 23, 38

Median = (22+23)/2 = 22.5

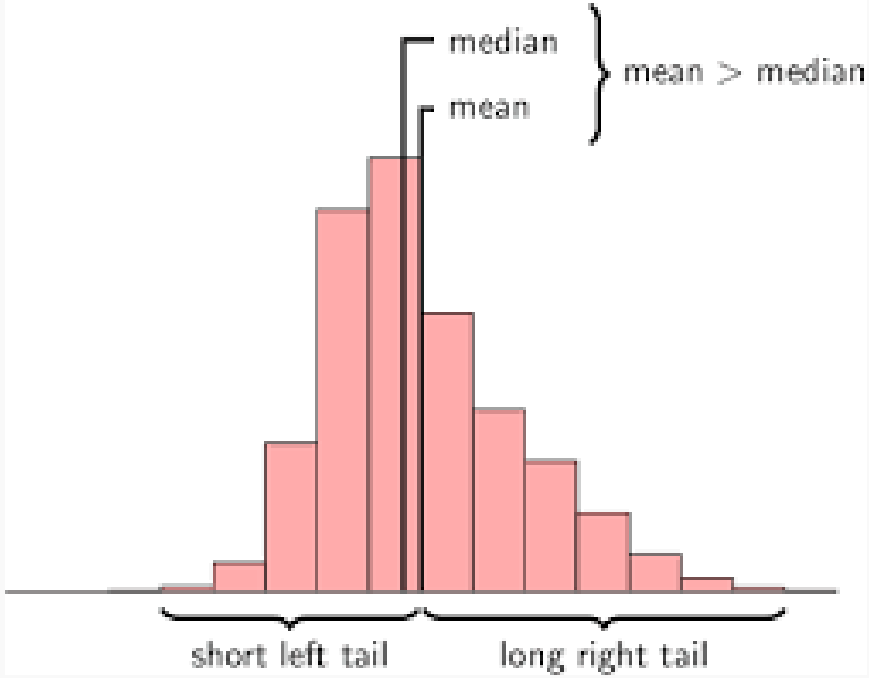
The median also describes what a typical observation looks like, or where is the center of the distribution of the sample of observations.

Mean vs. Median

The mean is sensitive to extreme values (**outliers**)

Mean, median, and skewness

The mean is sensitive to outliers:



The above distribution is called **right-skewed** since the mean is greater than the median. Note: **skewness** often "follows the longer tail".

Computational time

How hard (in terms of algorithmic complexity) is it to calculate

- the mean?

  at most $O(n)$

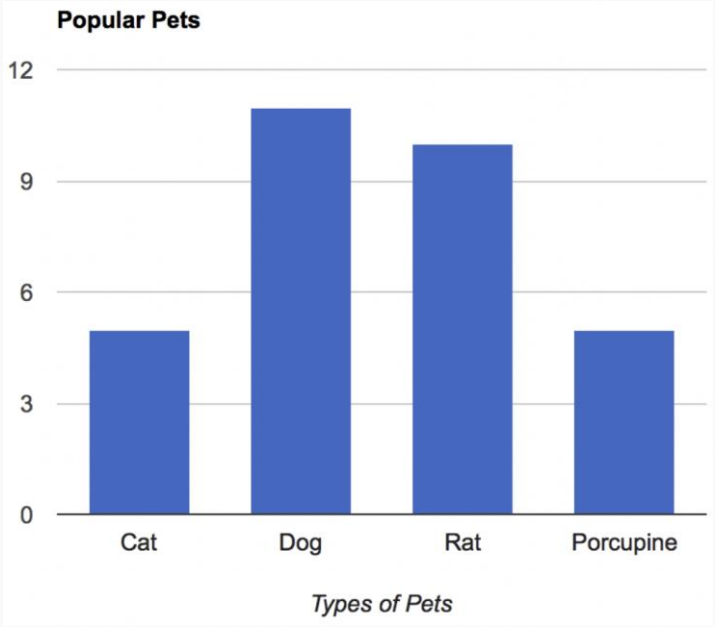
- the median?

  at most $O(n)$ or $O(n \log n)$

Note: Practicality of implementation should be considered!

Regarding Categorical Variables…

For categorical variables, neither mean or median make sense. Why?



The mode might be a better way to find the most "representative" value.

Measures of Spread: Range

The spread of a sample of observations measures how well the mean or median describes the sample.

One way to measure spread of a sample of observations is via the **range**.

Range = Maximum Value - Minimum Value

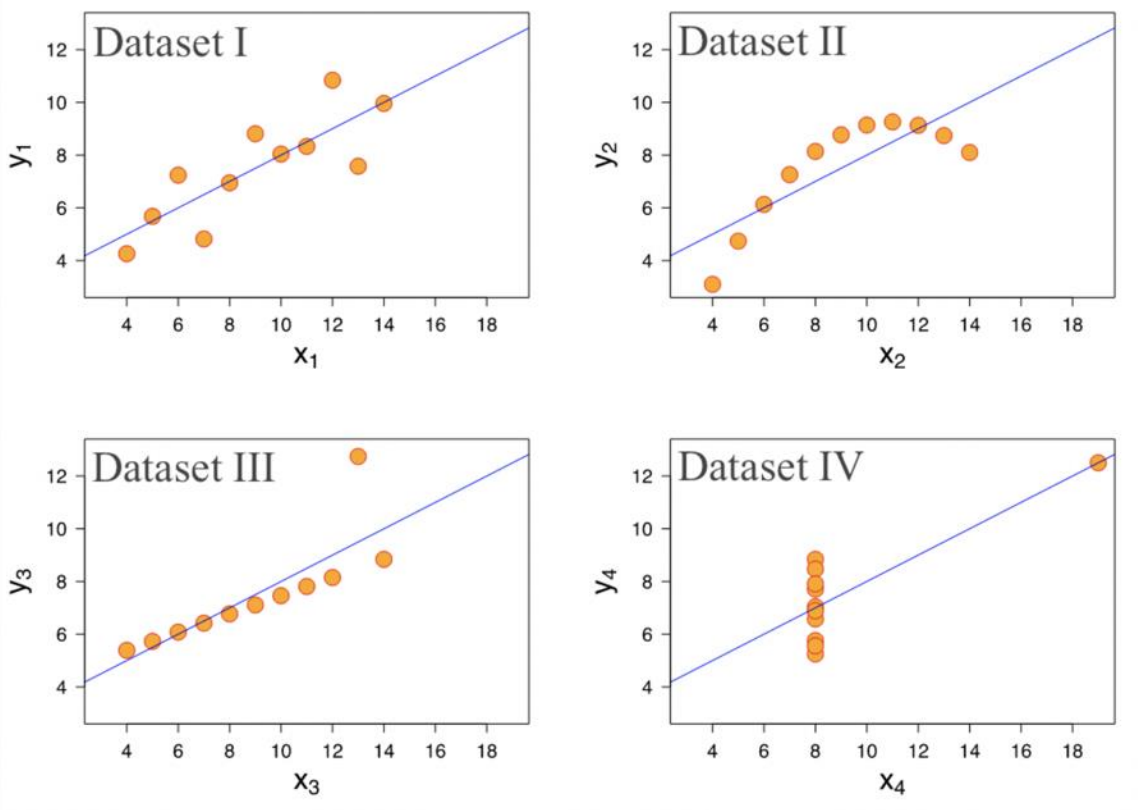
Anscombe's Data

The following four data sets comprise the Anscombe's Quartet; all four sets of data have identical simple summary statistics.

	Dataset I		Dataset II		Dataset III		Dataset IV	
	x	y	x	y	x	y	x	y
	10	8.04	10	9.14	10	7.46	8	6.58
	8	6.95	8	8.14	8	6.77	8	5.76
	13	7.58	13	8.74	13	12.74	8	7.71
	9	8.81	9	8.77	9	7.11	8	8.84
	11	8.33	11	9.26	11	7.81	8	8.47
	14	9.96	14	8.1	14	8.84	8	7.04
	6	7.24	6	6.13	6	6.08	8	5.25
	4	4.26	4	3.1	4	5.39	19	12.5
	12	10.84	12	9.13	12	8.15	8	5.56
	7	4.82	7	7.26	7	6.42	8	7.91
	5	5.68	5	4.74	5	5.73	8	6.89
Sum:	99.00	82.51	99.00	82.51	99.00	82.51	99.00	82.51
Avg:	9.00	7.50	9.00	7.50	9.00	7.50	9.00	7.50
Std:	3.32	2.03	3.32	2.03	3.32	2.03	3.32	2.03

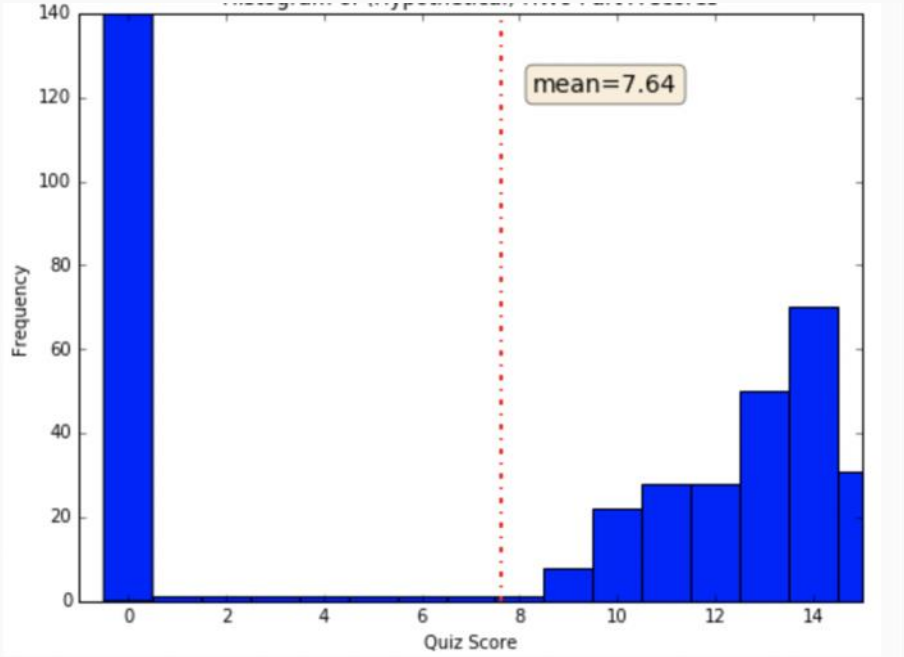
Anscombe's Data (cont.)

Summary statistics clearly don't tell the story of how they differ. But a picture can be worth a thousand words:



More Visualization Motivation

If I tell you that the average score for Homework 0 was: 7.64/15 = 50.9% last year, what does that suggest?



And what does the graph suggest?

More Visualization Motivation

Visualizations help us to analyze and explore the data. They help to:

- Identify hidden patterns and trends
- Formulate/test hypotheses
- Communicate any modeling results
  - Present information and ideas succinctly
  - Provide evidence and support
  - Influence and persuade
- Determine the next step in analysis/modeling

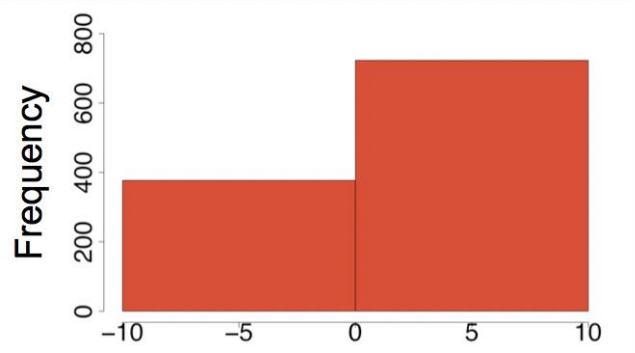
Types of Visualizations

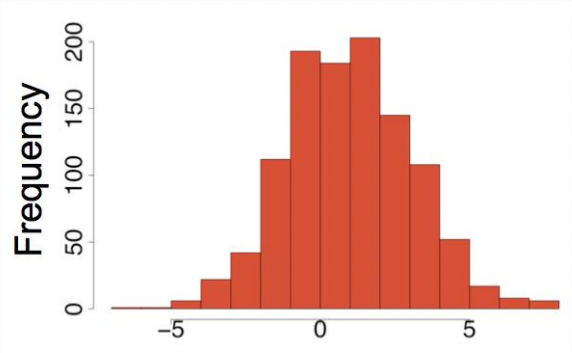
What do you want your visualization to show about your data?

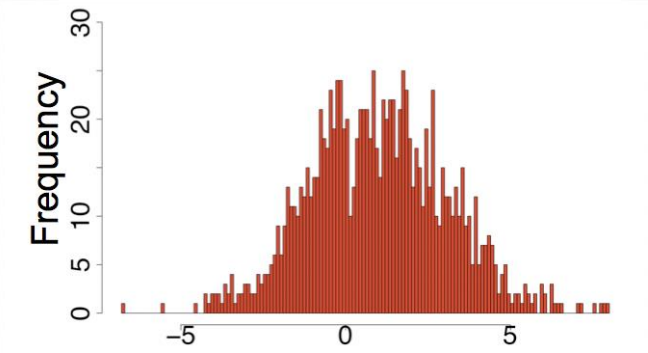
- **Distribution:** how a variable or variables in the dataset distribute over a range of possible values.
- **Relationship:** how the values of multiple variables in the dataset relate
- **Composition:** how the dataset breaks down into subgroups
- **Comparison:** how trends in multiple variable or datasets compare

Histograms to visualize distribution

A **histogram** is a way to visualize how 1-dimensional data is distributed across certain values.



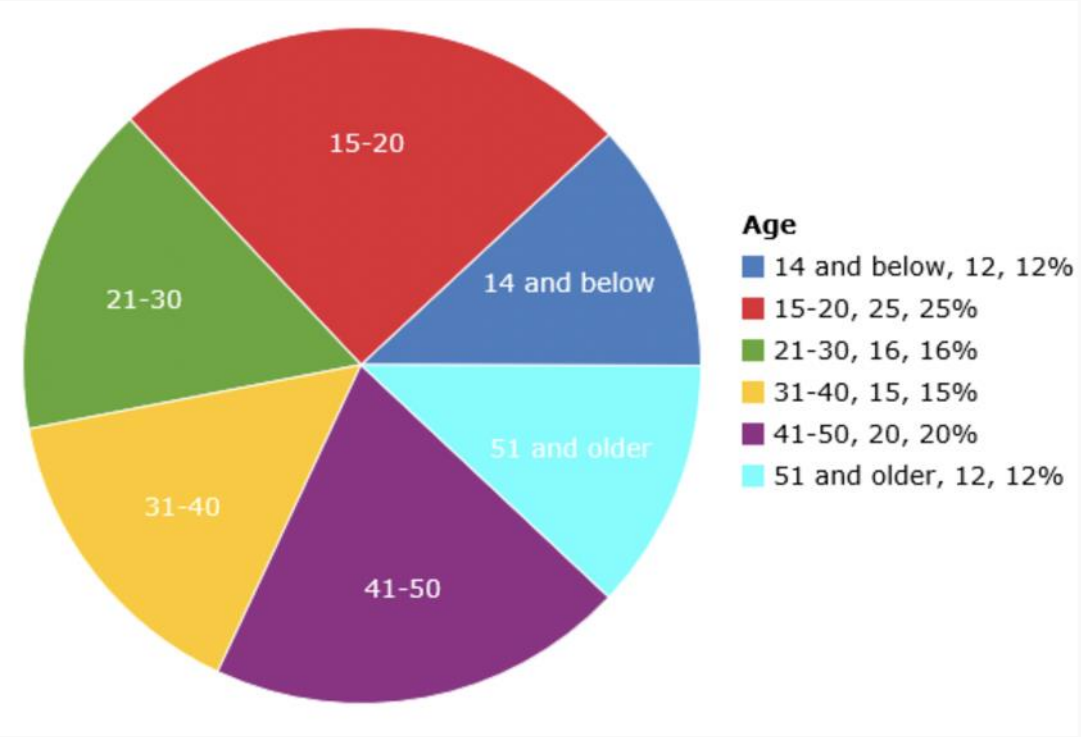




Note: Trends in histograms are sensitive to number of bins.

Pie chart for a categorical variable?

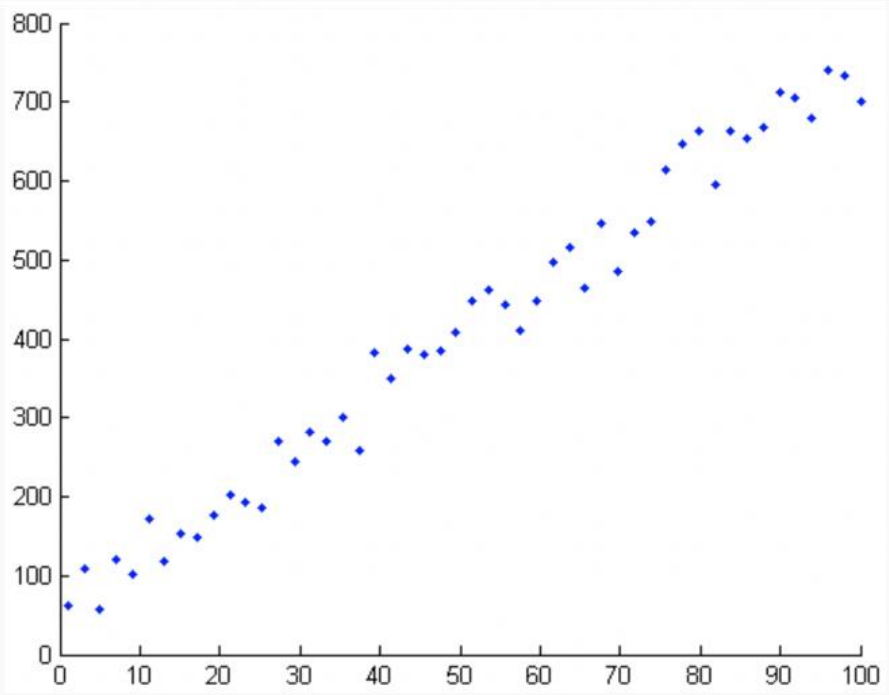
A **pie chart** is a way to visualize the static composition (aka, distribution) of a variable (or single group).

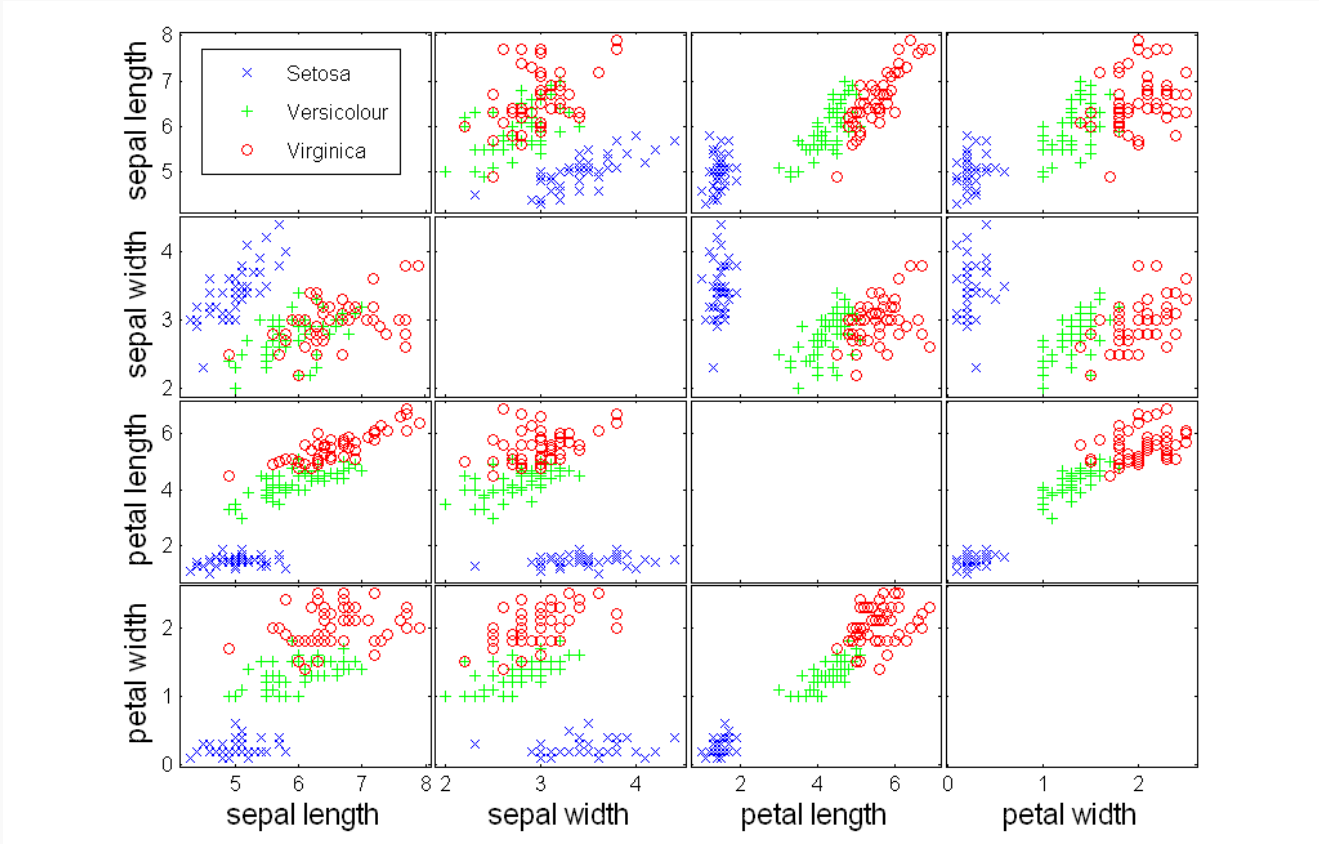


Pie charts are often frowned upon (and bar charts are used instead). Why?

Scatter plots to visualize relationships

A **scatter plot** is a way to visualize the relationship between two different attributes of multi-dimensional data.



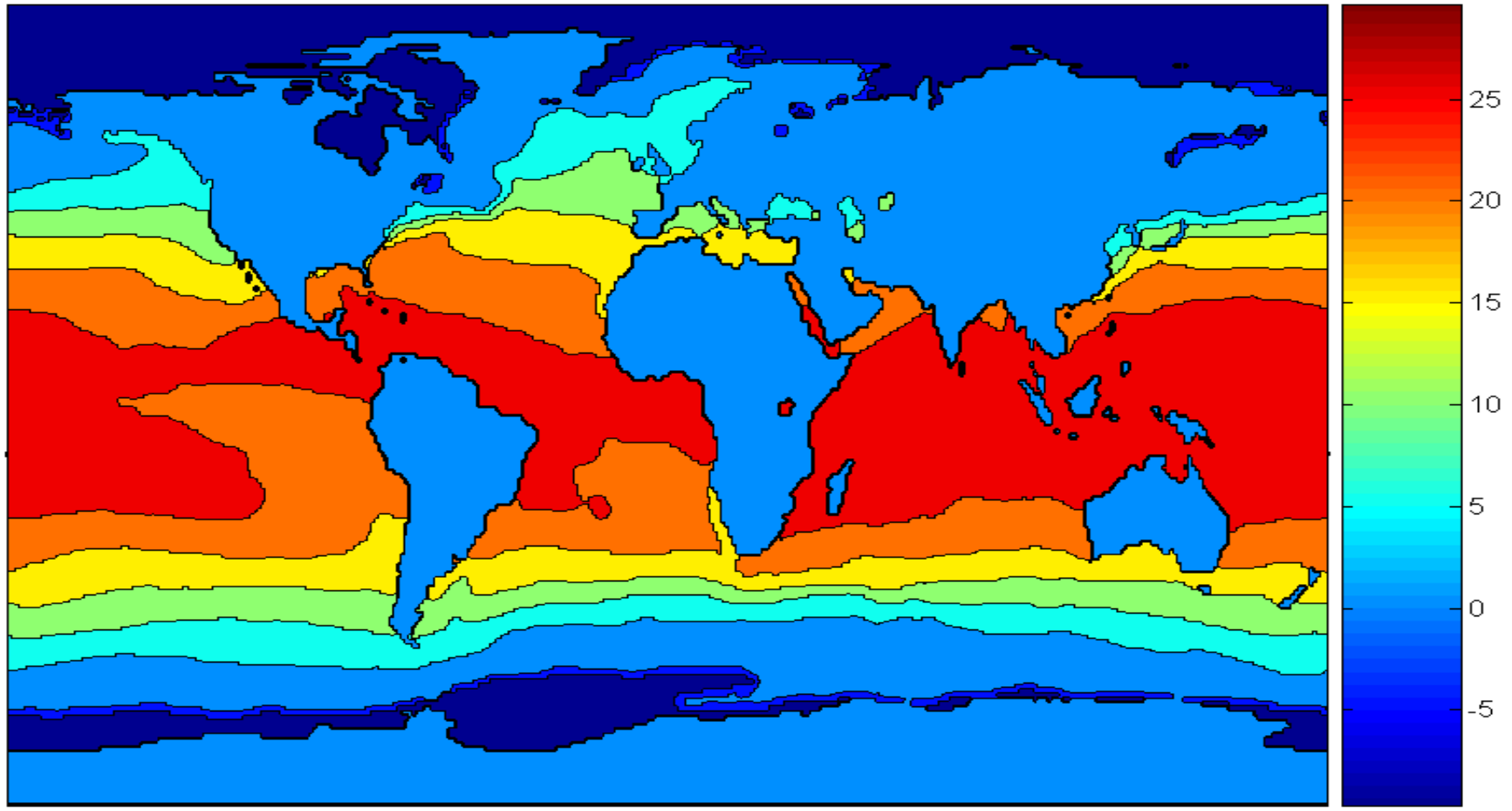


Visualization Techniques: Contour Plots

Contour plots

- Useful when a continuous attribute is measured on a spatial grid
- They partition the plane into regions of similar values
- The contour lines that form the boundaries of these regions connect points with equal values
- The most common example is contour maps of elevation
- Can also display temperature, rainfall, air pressure, etc.
  - An example for Sea Surface Temperature (SST) is provided on the next slide

Contour Plot Example: SST Dec, 1998

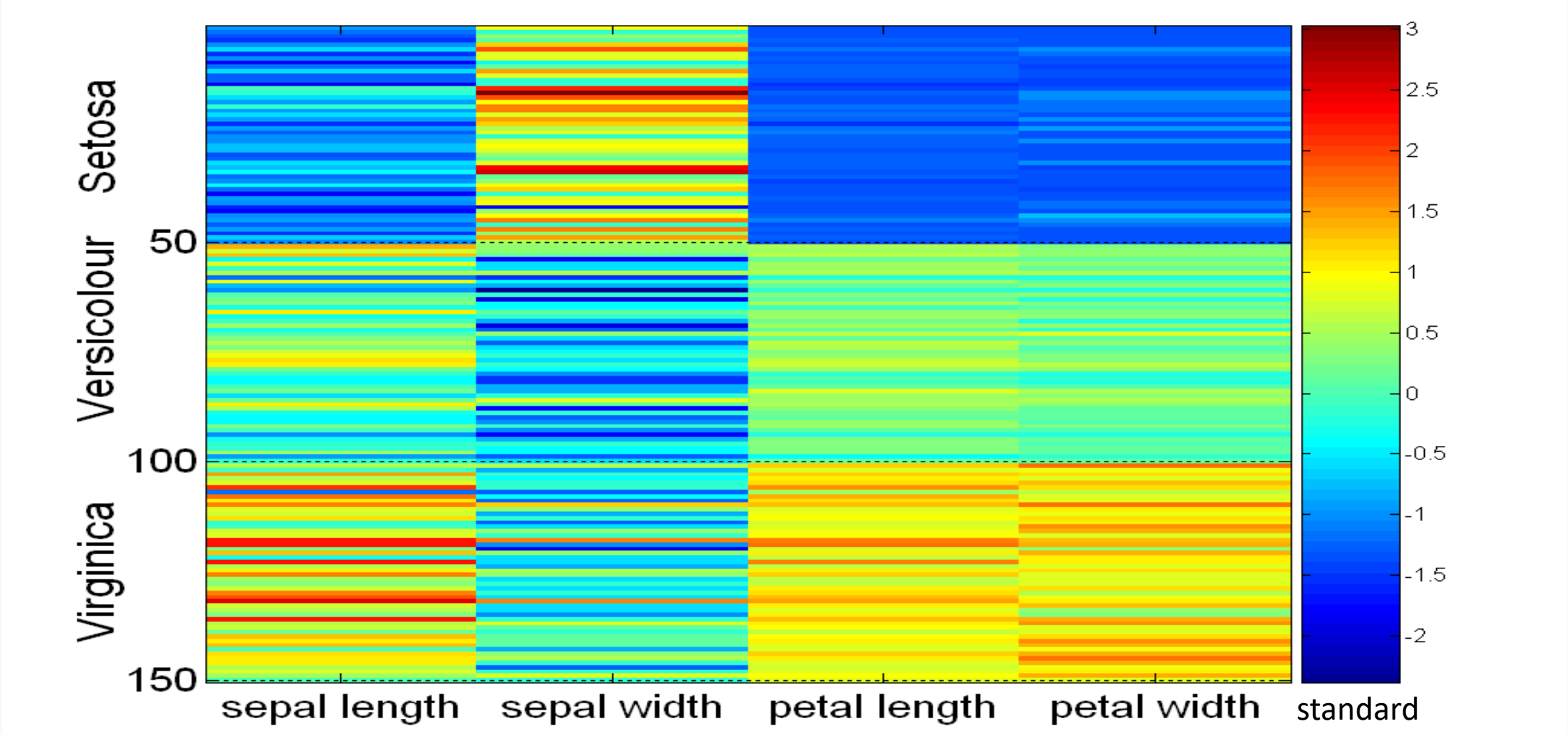


Visualization Techniques: Matrix Plots

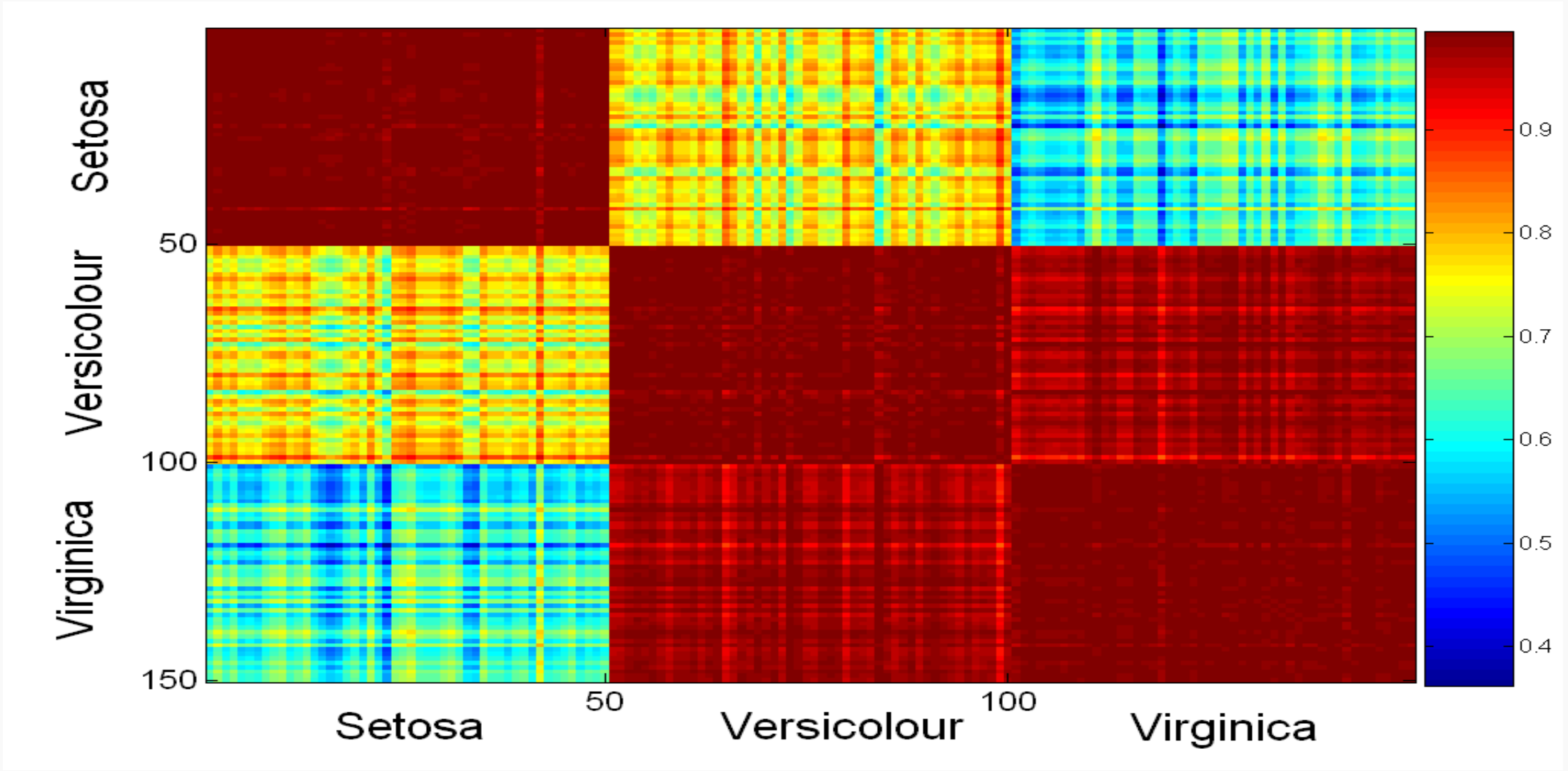
Matrix plots

- Can plot the data matrix
- This can be useful when objects are sorted according to class
- Typically, the attributes are normalized to prevent one attribute from dominating the plot
- Plots of similarity or distance matrices can also be useful for visualizing the relationships between objects
- Examples of matrix plots are presented on the next two slides

Visualization of the Iris Data Matrix



Visualization of the Iris Correlation Matrix

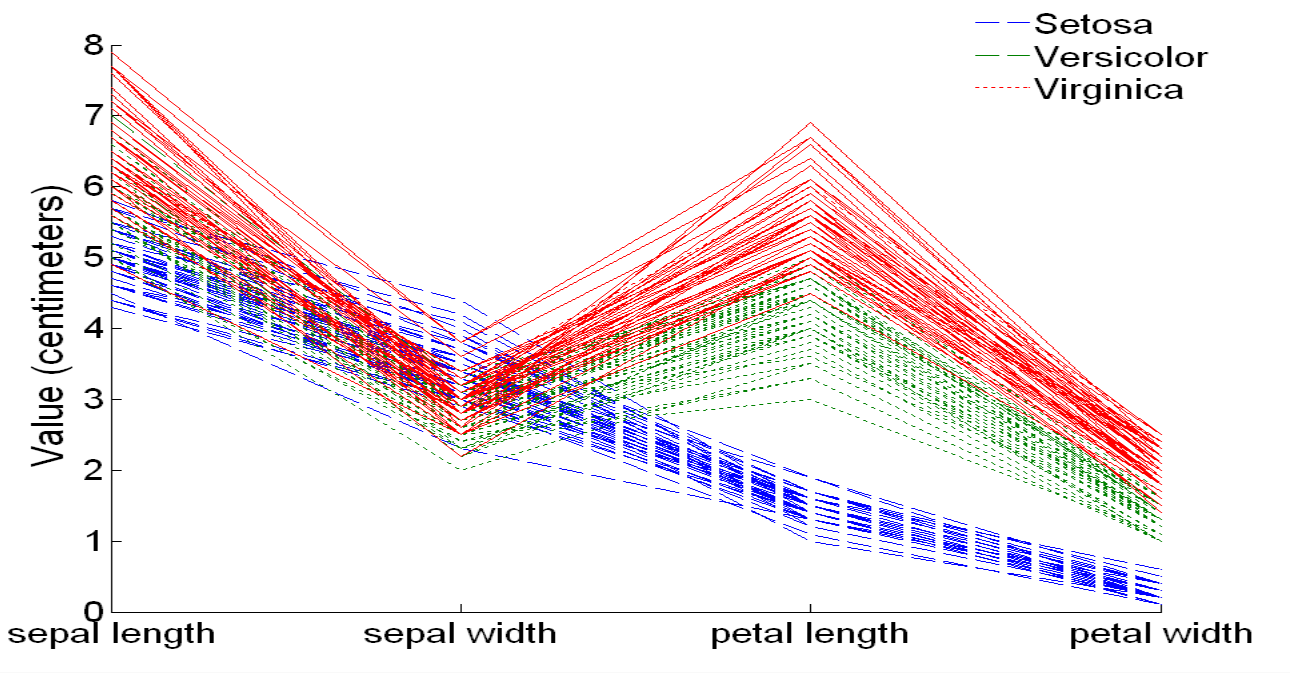


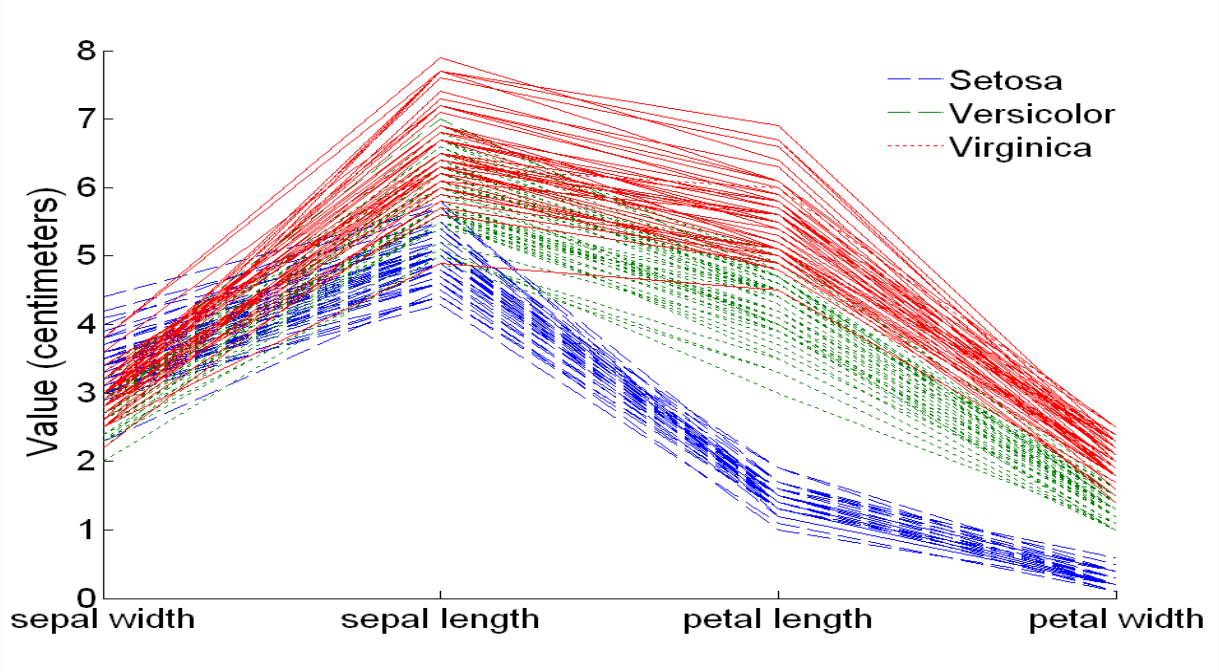
Visualization Techniques: Parallel Coordinates

Parallel Coordinates

- Used to plot the attribute values of high-dimensional data
- Instead of using perpendicular axes, use a set of parallel axes
- The attribute values of each object are plotted as a point on each corresponding coordinate axis and the points are connected by a line
- Thus, each object is represented as a line
- Often, the lines representing a distinct class of objects group together, at least for some attributes
- Ordering of attributes is important in seeing such groupings

Parallel Coordinates Plots for Iris Data





Other Visualization Techniques

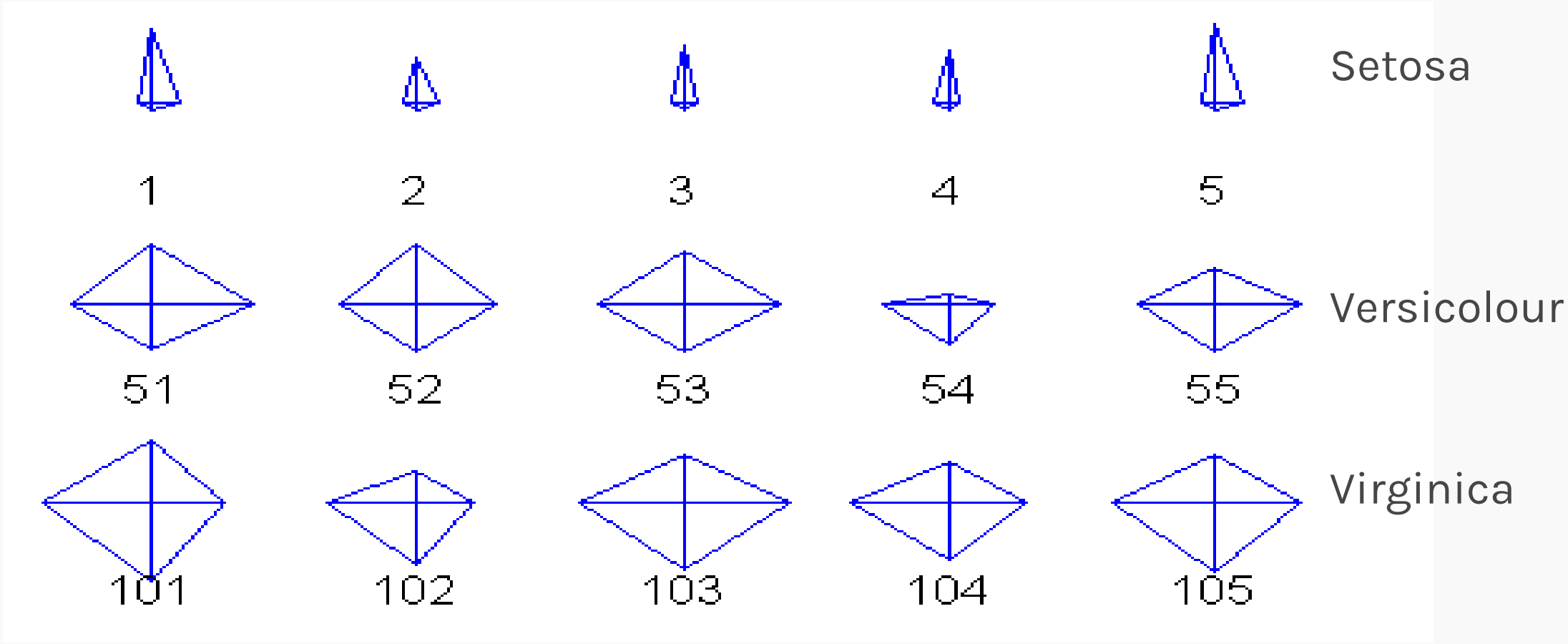
Star Plots

- Similar approach to parallel coordinates, but axes radiate from a central point
- The line connecting the values of an object is a polygon

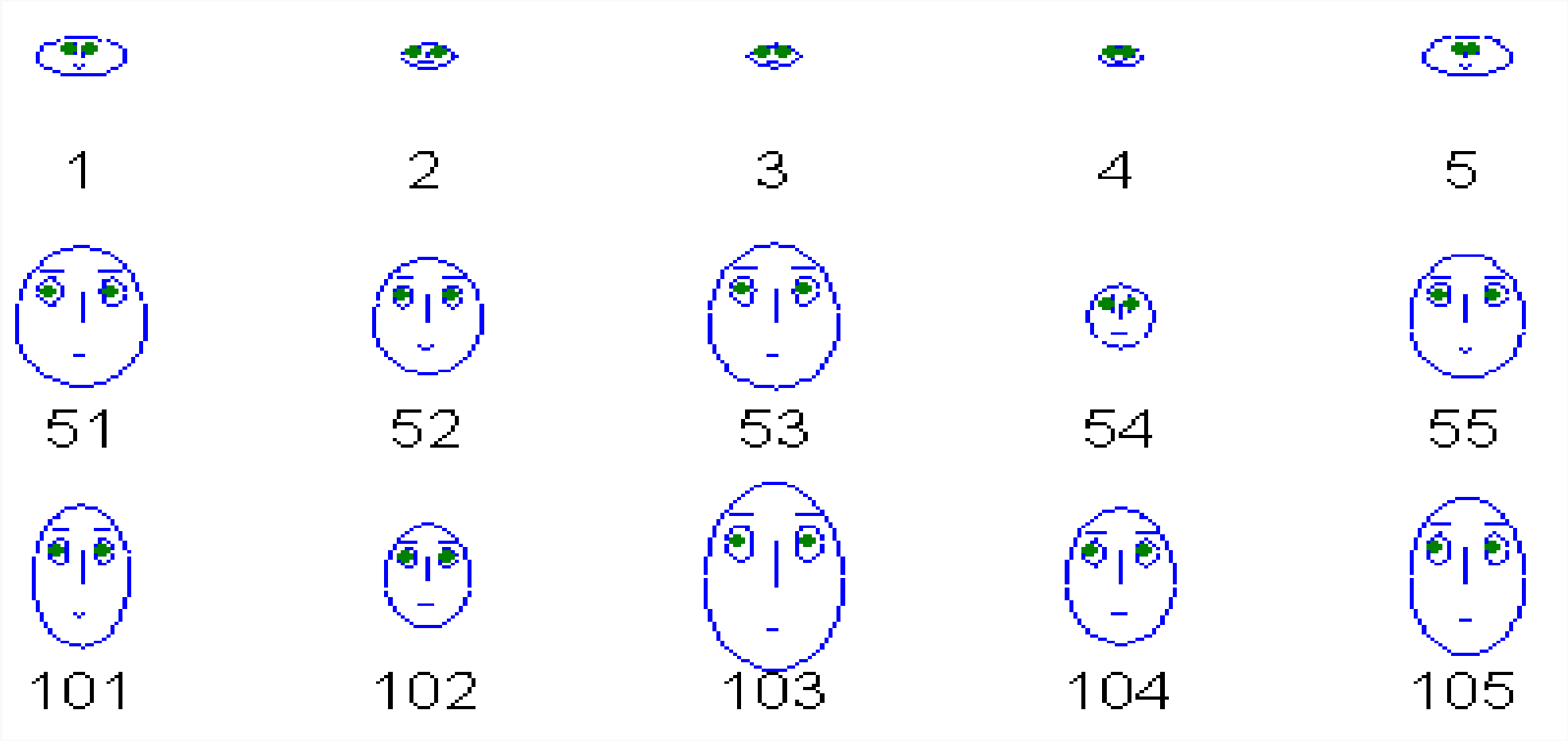
Chernoff Faces

- Approach created by Herman Chernoff
- This approach associates each attribute with a characteristic of a face
- The values of each attribute determine the appearance of the corresponding facial characteristic
- Each object becomes a separate face
- Relies on human's ability to distinguish faces

Star Plots for Iris Data

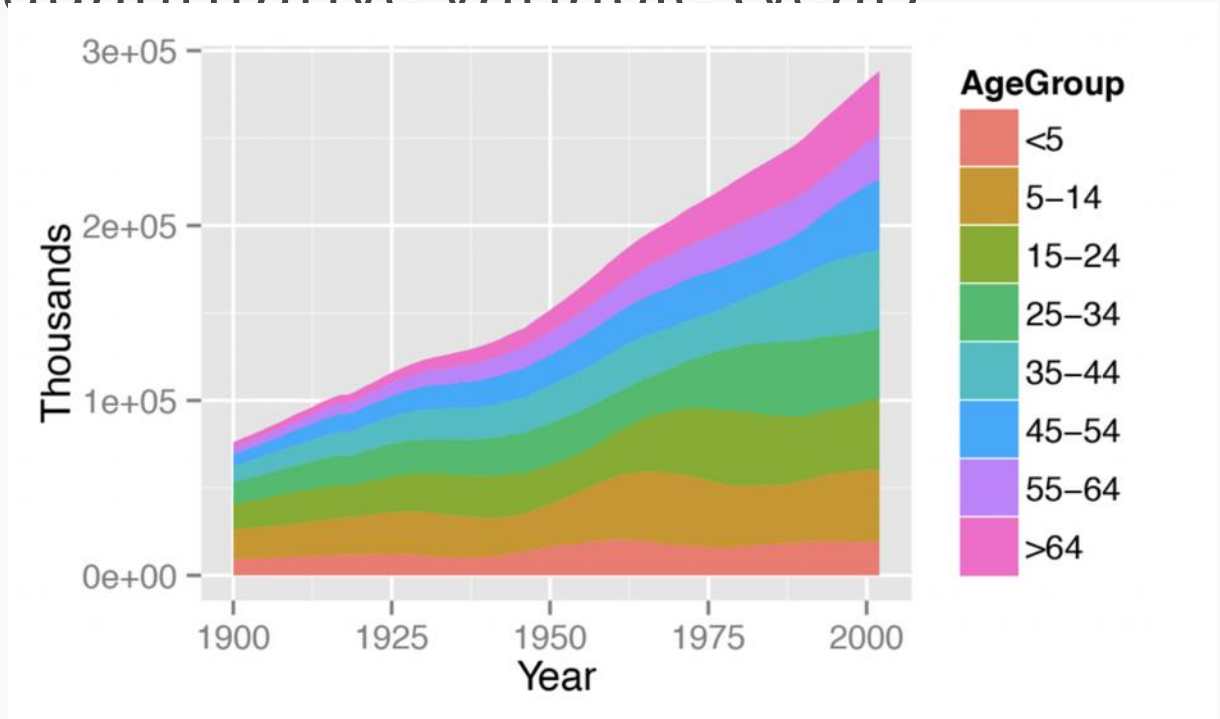


Chernoff Faces for Iris Data



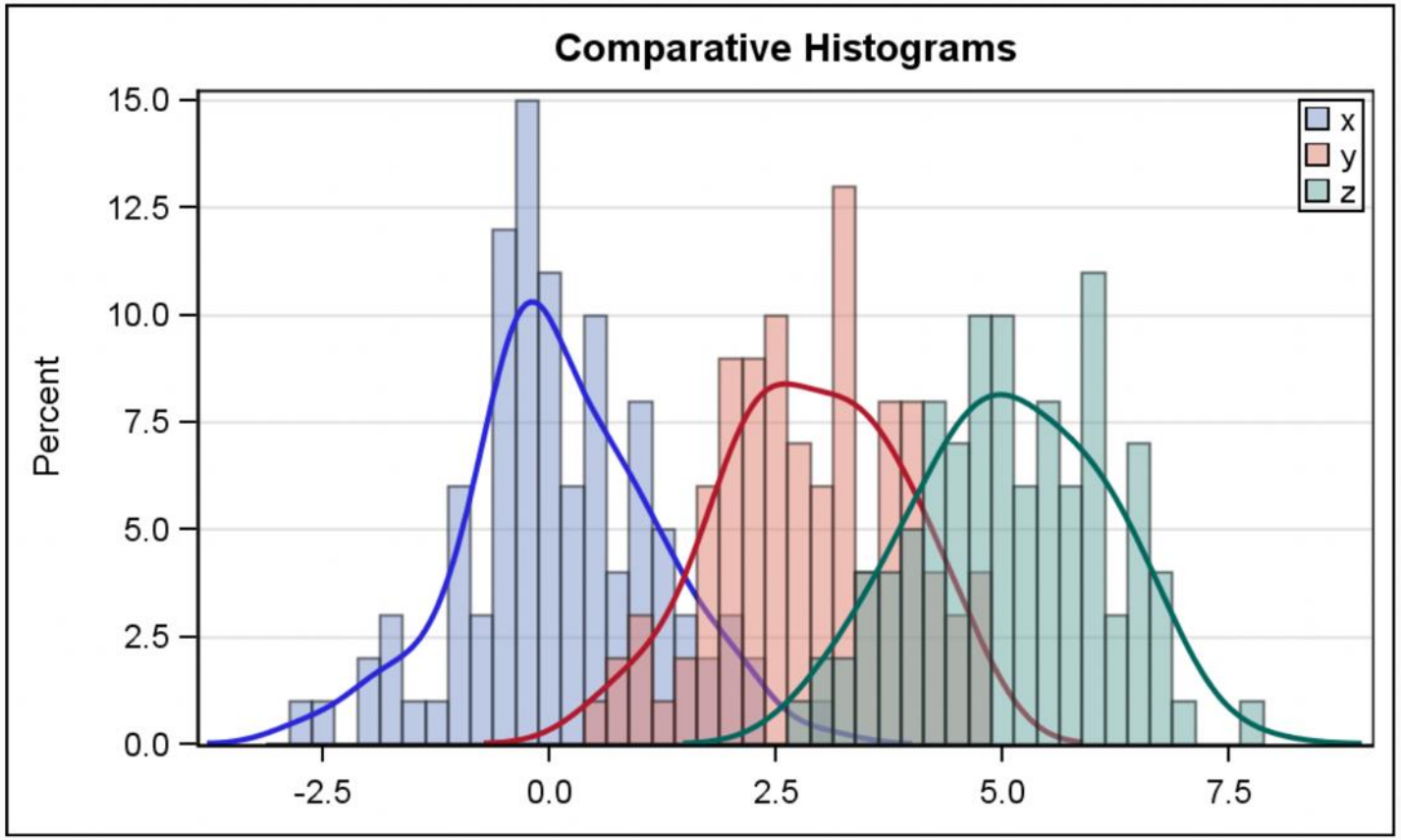
Stacked area graph to show trend over time

A **stacked area graph** is a way to visualize the composition of a group as it changes over time (or some other quantitative variable). This shows the relationship of a categorical variable (AgeGroup) to a quantitative variable (year)



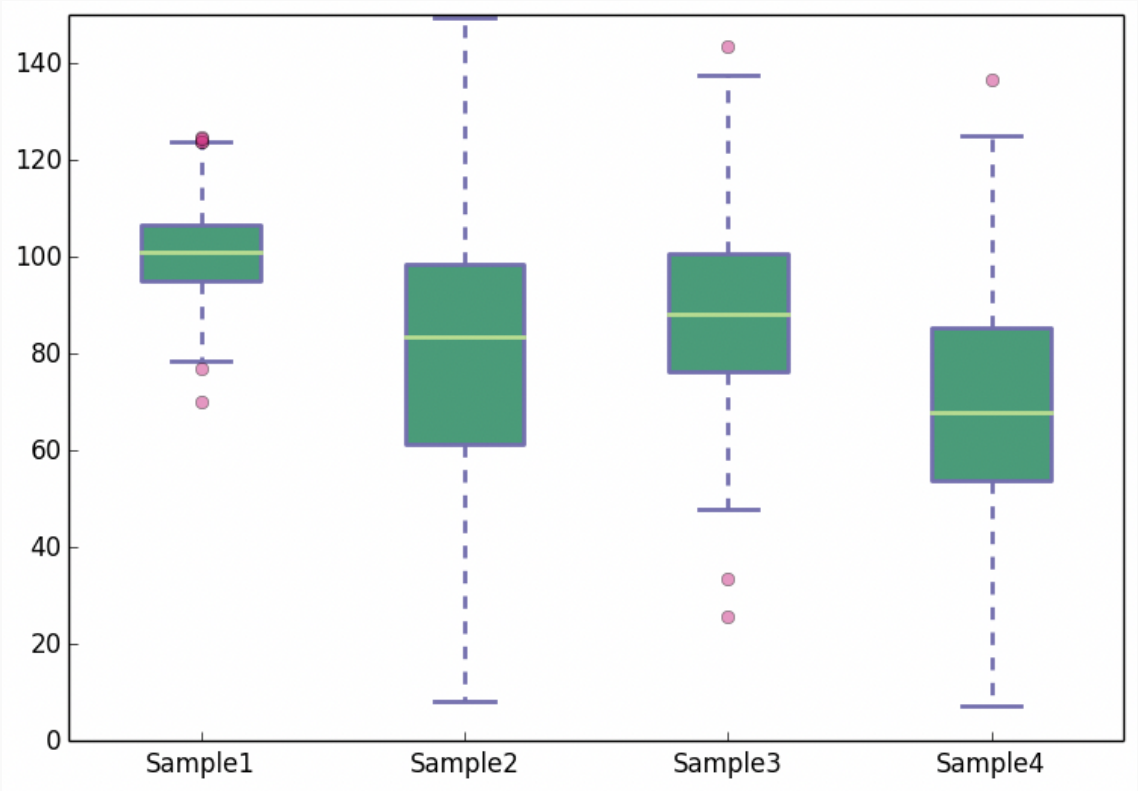
Multiple histograms

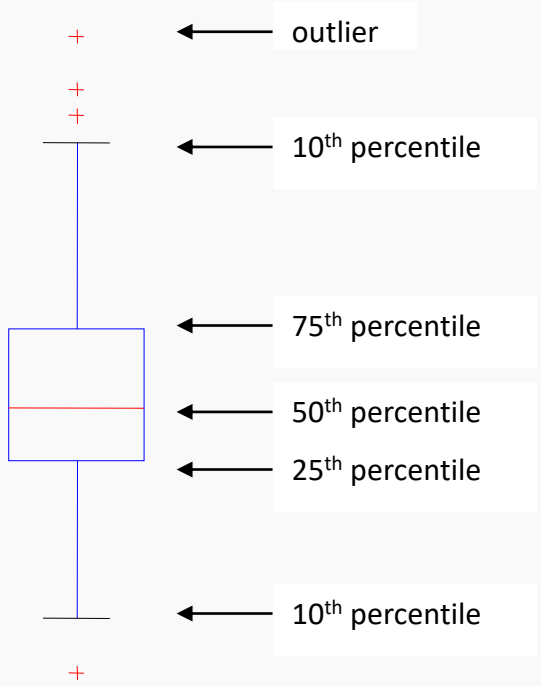
Plotting **multiple histograms** (and **kernel density estimates** of the distribution, here) on the same axes is a way to visualize how different variables compare (or how a variable differs over specific group



Boxplots

A **boxplot** is a simplified visualization to compare a quantitative variable across groups. It highlights the range, quartiles, median and any outliers present in a data set.





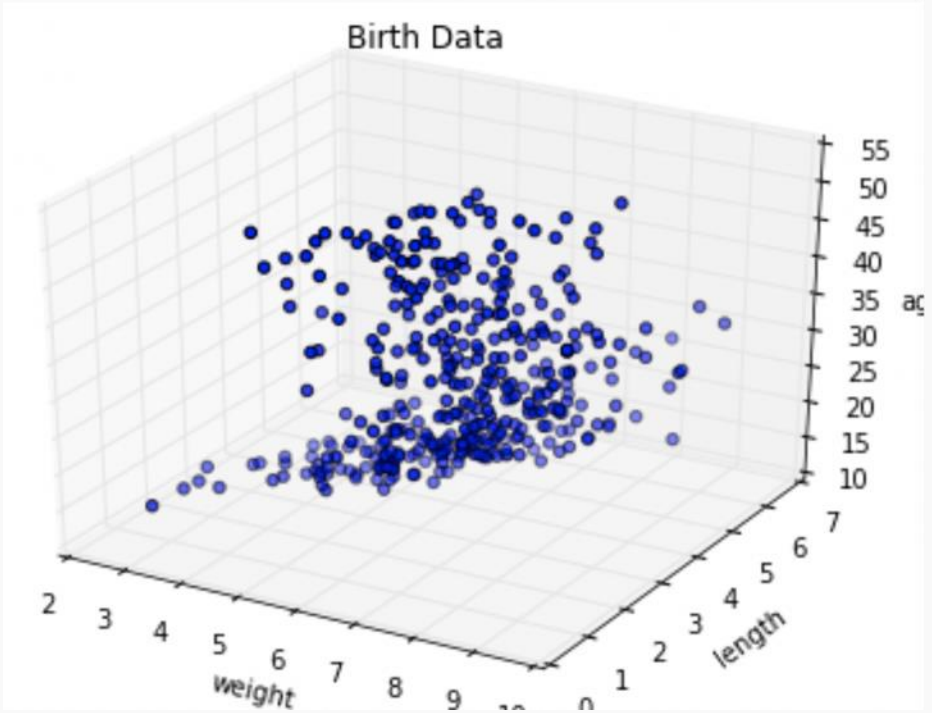
[Not] Anything is possible!

Often your dataset seem too complex to visualize:

- Data is too high dimensional (how do you plot 100 variables on the same set of axes?)
- Some variables are categorical (how do you plot values like Cat or No?)

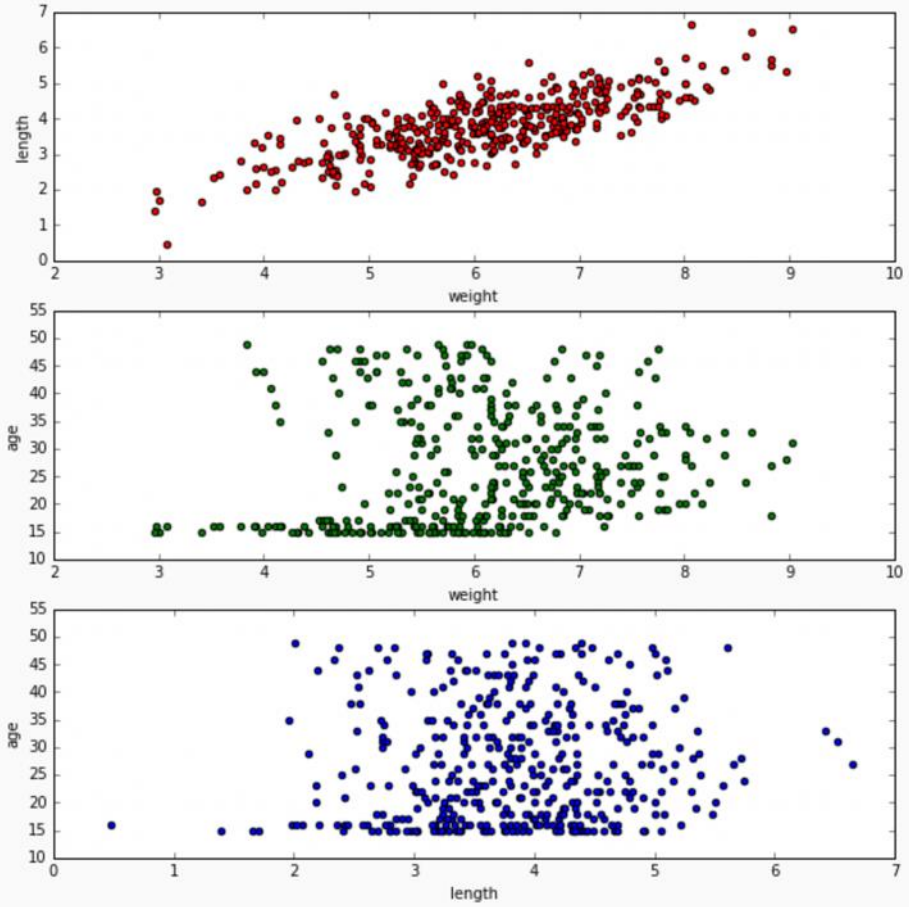
More dimensions not always better

When the data is high dimensional, a scatter plot of all data attributes can be impossible or unhelpful



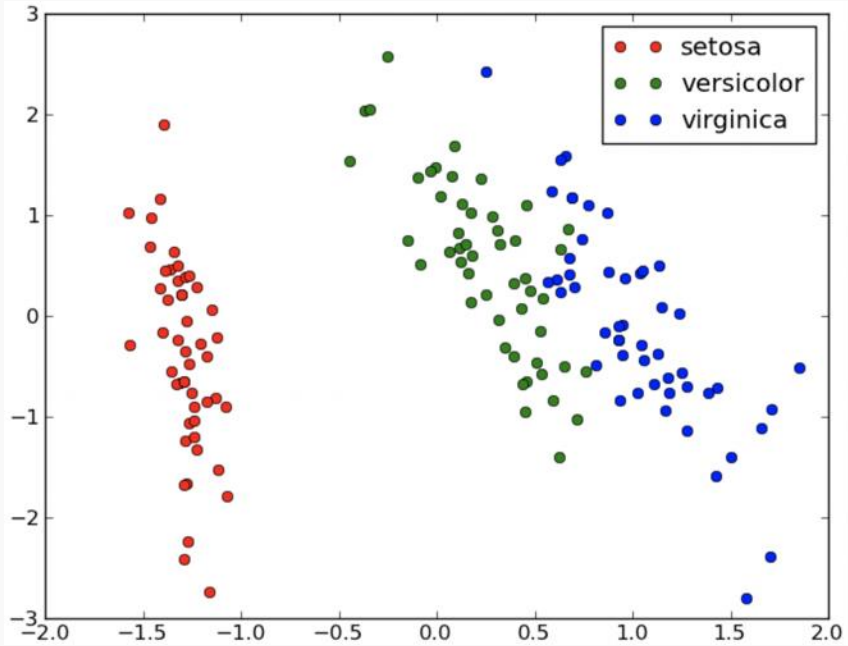
Reducing complexity

Relationships may be easier to spot by producing multiple plots of lower dimensionality



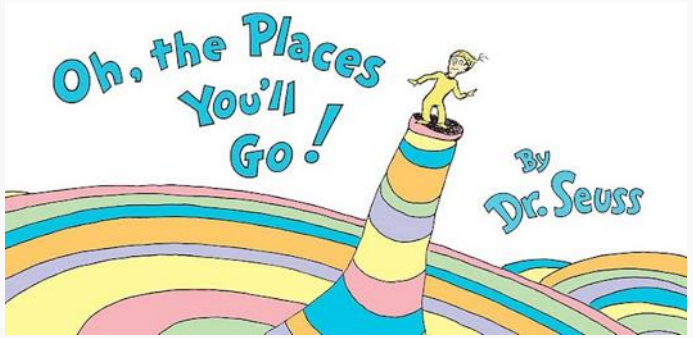
Reducing complexity

For 3D data, color coding a categorical attribute can be "effective"



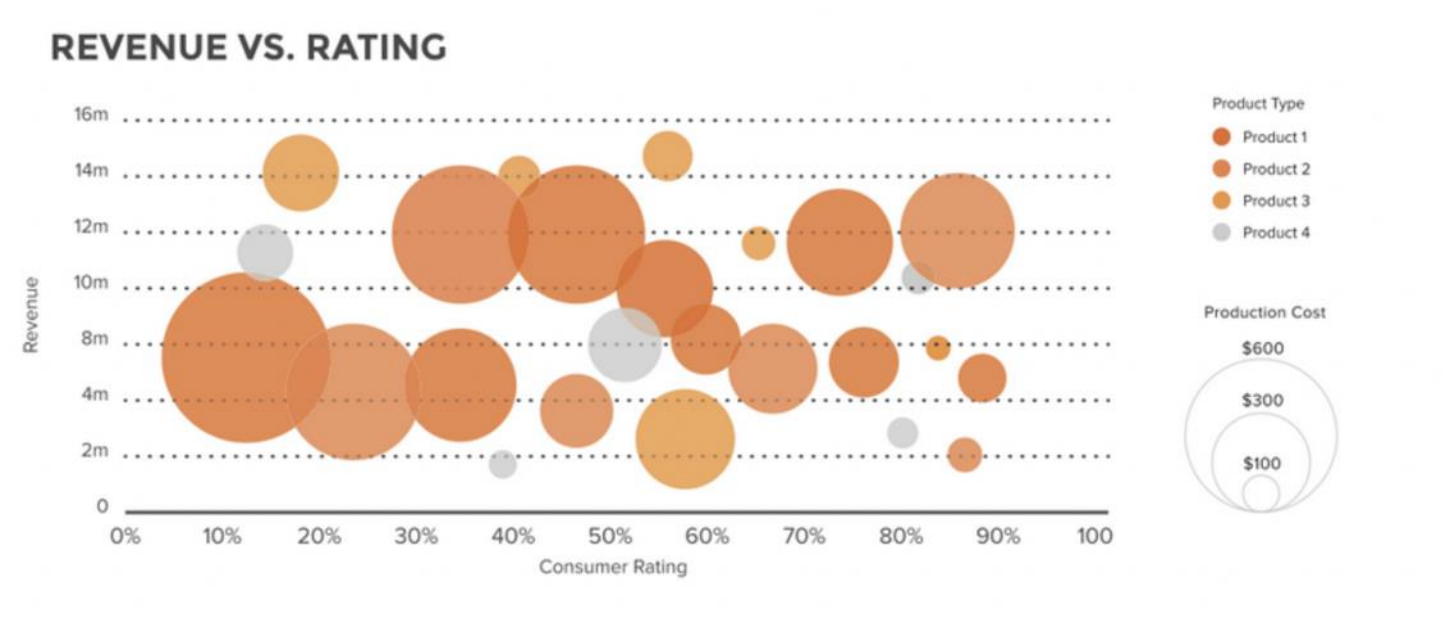
This visualizes a set of Iris measurements. The variables are: petal length, sepal length, Iris type (setosa, versicolor, virginica).

Except when it's not effective.
What could be a better choice?



3D can work

For 3D data, a quantitative attribute can be encoded by size in a bubble chart

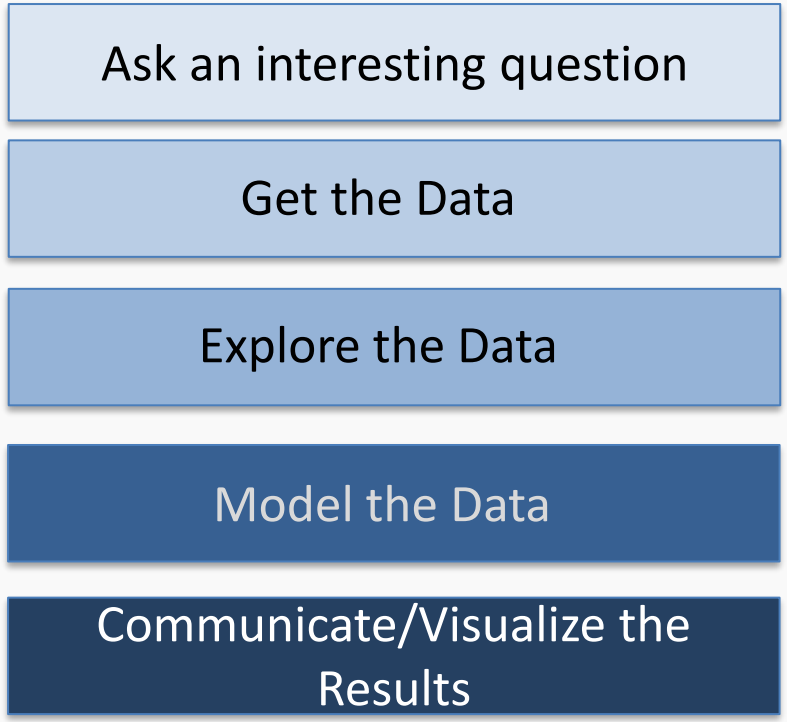


The above visualizes a set of consumer products. The variables are: revenue, consumer rating, product type and product cost.

An Example: The Data Science Process

What?

The Data Science Process is similar to the scientific process - one of observation, model building, analysis, and conclusion:



Note: This process is by no means linear!

Analyzing Hubway Data

Introduction: Hubway is metro-Boston's public bike share program, with more than 1600 bikes at 160+ stations across the Greater Boston area. Hubway is owned by four municipalities in the area.

By 2016, Hubway operated 185 stations and 1750 bicycles, with 5 million ride since launching in 2011.

The Data: In April 2017, Hubway held a Data Visualization Challenge at the Microsoft NERD Center in Cambridge, releasing 5 years of trip data.

The Question: What does the data tell us about the ride share program?

The Data Exploration/Question Refinement Cycle

Our original question: **'What does the data tell us about the ride share program?'** is a reasonable slogan to promote a hackathon. It is not good for guiding scientific investigation.

Before we can refine the question, we have to look at the data!

	seq_id	hubway_id	status	duration	start_date	strt_statn	end_date	end_statn	bike_nr	subsc_type	zip_code	birth_date	gender
0	1	8	Closed	9	7/28/2011 10:12:00	23.0	7/28/2011 10:12:00	23.0	B00468	Registered	'97217	1976.0	Male
1	2	9	Closed	220	7/28/2011 10:21:00	23.0	7/28/2011 10:25:00	23.0	B00554	Registered	'02215	1966.0	Male
2	3	10	Closed	56	7/28/2011 10:33:00	23.0	7/28/2011 10:34:00	23.0	B00456	Registered	'02108	1943.0	Male
3	4	11	Closed	64	7/28/2011 10:35:00	23.0	7/28/2011 10:36:00	23.0	B00554	Registered	'02116	1981.0	Female
4	5	12	Closed	12	7/28/2011 10:37:00	23.0	7/28/2011 10:37:00	23.0	B00554	Registered	'97214	1983.0	Female

Based on the data, what kind of questions can we ask?

The Data Exploration/Question Refinement Cycle

Who? Who's using the bikes?

Refine into specific hypotheses:

- More men or more women?
- Older or younger people?
- Subscribers or one time users?

The Data Exploration/Question Refinement Cycle

Where? Where are bikes being checked out?

Refine into specific hypotheses:

- More in Boston than Cambridge?
- More in commercial or residential?
- More around tourist attractions?

Sometimes the data is given to you in pieces and must be merged!

The Data Exploration/Question Refinement Cycle

When? When are the bikes being checked out?

Refine into specific hypotheses:

- More during the weekend than on the weekdays?
- More during rush hour?
- More during the summer than the fall?

Sometimes the feature you want to explore doesn't exist in the data, and must be engineered!

The Data Exploration/Question Refinement Cycle

Why? For what reasons/activities are people checking out bikes?

Refine into specific hypotheses:

- More bikes are used for recreation than commute?
- More bikes are used for touristic purposes?
- Bikes are use to bypass traffic?

Do we have the data to answer these questions with reasonable certainty?

What data do we need to collect in order to answer these questions?

The Data Exploration/Question Refinement Cycle

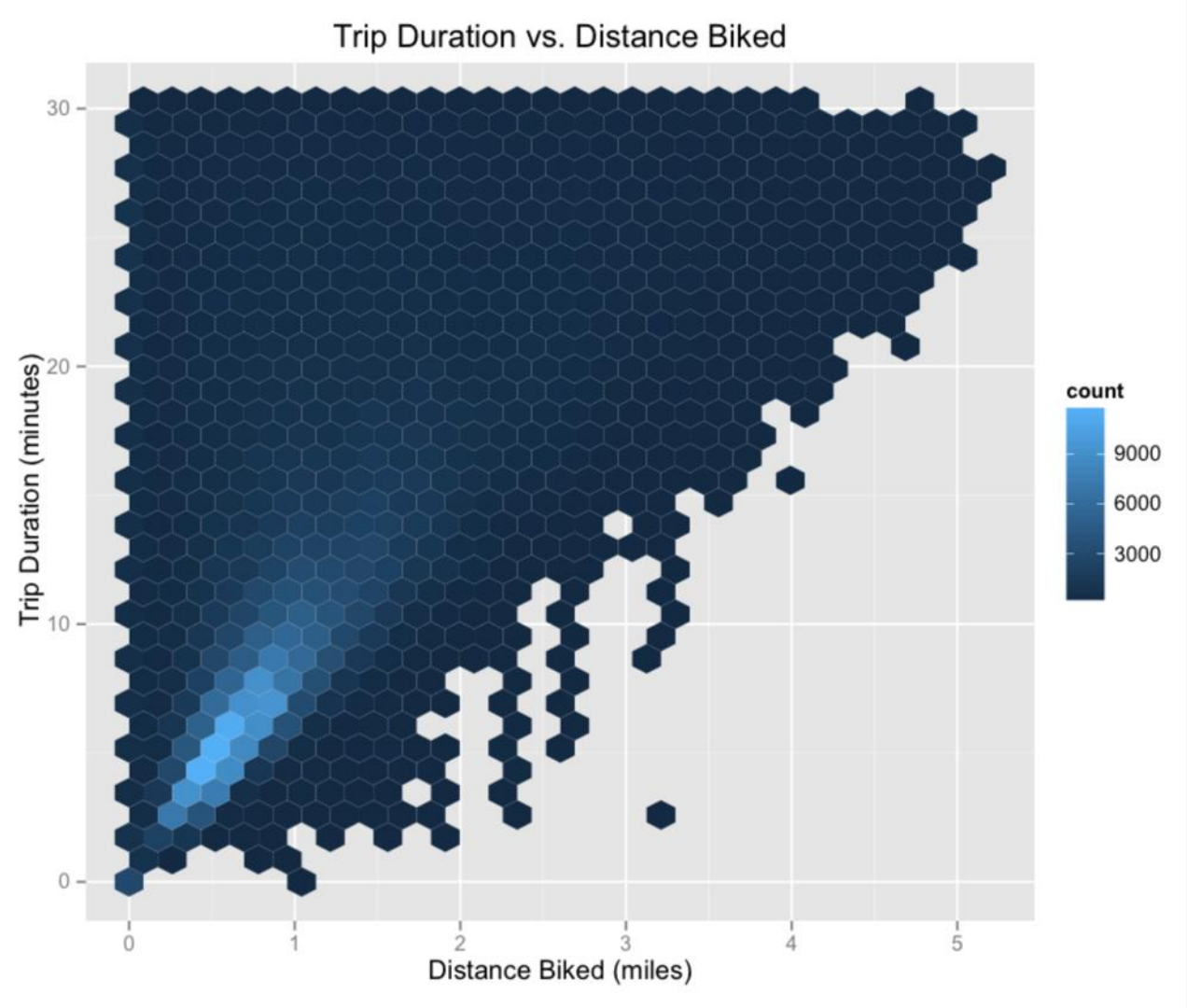
How? Questions that combine variables.

- How does user demographics impact the duration the bikes are being used? Or where they are being checked out?
- How does weather or traffic conditions impact bike usage?
- How do the characteristics of the station location affect the number of bikes being checked out?

How questions are about modeling relationships between different variables.

Inspirations for Data Viz/Exploration

So how well did we do in formulating creative hypotheses and manipulating the data for answers?

Check out the winners of the Hubway Challenge:

http://hubwaydatachallenge.org